Polynomial Boundary Coordinates for Collatz Cycles

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Abstract

This paper studies arithmetic obstructions to realizing prescribed parity patterns as positive integer cycles of the shortcut Collatz map. A binary parity word of length n with u odd symbols determines a rational periodic orbit, but the standard numerator formula does not reveal how the arrangement of those symbols affects integrality. Toward characterizing this dependence, we develop coordinates that measure a prescribed parity pattern relative to the corresponding upper-Christoffel mechanical word. The construction is developed in the positive cycle-relevant range $u < n < 2u$ and $2^n > 3^u$. Under the additional condition $\gcd(n, u) = 1$, the map $t \mapsto (n - u)t \bmod u$ permutes the u odd phases and places them in a lossless residue order. Each cyclic parity class then has a unique rotation whose cumulative zero path lies below the balanced mechanical path of the corresponding upper-Christoffel representative. The nonnegative vertical displacement between these paths is called the *bridge* L. Writing its heights in residue order and taking a discrete boundary produces a sparse polynomial Q_L whose occupied coefficients are exactly the $K(L)$ plateaux of the reordered bridge.

The rational orbit is integral exactly when Q_L is divisible by $2X^{n-u} - 3$ in $\mathbb{Z}[X]/(X^u - 2)$. When this holds, the profile quotient recovers the odd orbit and the boundary quotient has full support. Integral realizability therefore turns a sparse geometric boundary into a dense arithmetic response. Only the $K(L)$ occupied coefficients of Q_L can interrupt the induced quotient recurrence: a long intervening gap forces exponential 3-adic growth, while bridge geometry bounds the sum of the absolute boundary coefficients in terms of $K(L)$ and Laurent's estimate controls the denominator loss. Balancing these constraints gives an absolute constant $C > 0$ such that every hypothetical coprime positive cycle satisfies $K(L) \geq \sqrt{\log_2 3}\,\sqrt{u} - C(1 + \log u)^2$. Thus such a cycle cannot lie in the low-boundary-complexity regime below the square-root scale.

Finally, the analysis recovers Knight's high-cycle exclusion and excludes every one-swap mechanical word from encoding a nontrivial positive integer Collatz cycle.

AI provenance and disclosure. This is human-directed, AI-generated mathematical research. The author supplied an early recurrence relation and parity-word explorations; all subsequent mathematical development and manuscript production were AI-generated under his direction, selection, and editorial review. A fuller account of system roles, verification status, and responsibility appears in Appendix B.

1 Introduction

1.1 Prescribed parity cycles and integral realizability

The shortcut Collatz map sends a positive integer x to $x/2$ when x is even and to $(3x + 1)/2$ when x is odd. The conjecture that every positive integer eventually reaches 1 remains open; in particular, no nontrivial positive cycle—a periodic orbit other than $(1, 2)$—is known to exist. This paper studies the constraints that the parity pattern of any such cycle must satisfy.

To prescribe parity patterns before integrality is known, we extend the Collatz map to the rationals with odd denominator:

$$\mathbb{Z}_{(2)} := \left\{ \frac{a}{b} \in \mathbb{Q} : a, b \in \mathbb{Z},\ b \neq 0,\ b \text{ odd} \right\}. \tag{1}$$

This is the localization of $\mathbb{Z}$ at the prime 2, conventionally denoted $\mathbb{Z}_{(2)}$. For $\varepsilon \in \{0, 1\}$, let

$$T_0(x) := \frac{x}{2}, \qquad T_1(x) := \frac{3x + 1}{2}. \tag{2}$$

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If $x = a/b \in \mathbb{Z}_{(2)}$ is written in lowest terms, define its parity to be the parity of a, and define

$$T(x) := T_{a \bmod 2}(x). \tag{3}$$

This map preserves $\mathbb{Z}_{(2)}$ and restricts on the integers to the shortcut Collatz map.

Now prescribe the successive branches by a word

$$w = (w_0, \dots, w_{n-1}) \in \{0,1\}^n, \qquad |w|_1 := \sum_{j=0}^{n-1} w_j = u, \qquad z := n - u, \tag{4}$$

and define

$$T_w := T_{w_{n-1}} \circ \cdots \circ T_{w_0}. \tag{5}$$

A standard expansion gives

$$2^n T_w(x) = 3^u x + N(w), \tag{6}$$

where

$$N(w) := \sum_{j=0}^{n-1} w_j 2^j 3^{\sum_{m=j+1}^{n-1} w_m}. \tag{7}$$

Thus, with

$$D := 2^n - 3^u, \tag{8}$$

the affine map T_w has the unique rational fixed point

$$x(w) = \frac{N(w)}{D}. \tag{9}$$

Since a positive power of 2 cannot equal a power of 3, one has $D \neq 0$. Moreover, D is odd, so $x(w) \in \mathbb{Z}_{(2)}$. The prescribed-parity realization theorem proved in Section 2.1 shows that the successive parities of this orbit are precisely the symbols of w. When $D > 0$ and $u > 0$, the word determines a positive integral periodic orbit exactly when

$$D \mid N(w). \tag{10}$$

In this case w is **integrally realizable**. Its displayed length n need not be its least period. Cyclic rotation changes the numerator $N(w)$ but only shifts the initial point on the associated rational orbit; integrality is therefore a property of the **cyclic parity class** of w,

$$[w]_{\mathrm{cyc}} := \{(w_{j+k \bmod n})_{j=0}^{n-1} : 0 \le k < n\}, \tag{11}$$

rather than of a chosen linear representative.

The prescribed-parity formula, originating with Böhm and Sontacchi [1] and developed systematically in the study of rational Collatz cycles by Lagarias [2], solves the rational realization problem. It does not, however, organize the integral one. The numerator $N(w)$ is a nonlocal mixed-base sum: the contribution of each odd symbol depends both on its temporal position and on the number of later odd symbols, and these dependences occur through different exponential bases. The formula also depends on a chosen cyclic root, although its divisibility by D does not. Consequently, the condition $D \mid N(w)$ reveals little directly about the internal geometry of the parity pattern.

The problem addressed here is to test this divisibility within coordinates that expose the structure of the cyclic parity class. The construction selects a canonical odd root, measures displacement from a balanced reference pattern, reorganizes the resulting heights in an arithmetic residue order, and compresses them to a sparse but lossless polynomial boundary Q_L. Its purpose is arithmetic: integral realizability becomes a factorization $Q_L = (2X^{n-u} - 3)C_L$ in $\mathbb{Z}[X]/(X^u - 2)$. The boundary has only $K(L)$ occupied coefficients, whereas the quotient of a least-period realization has no zero coefficient. The resulting tension between sparse geometric forcing and a dense arithmetic response drives the plateau lower bound.

The present work concerns only nontrivial positive cycles. Section 2.1 proves that a nontrivial positive cycle of least period n, with u odd members, lies in the strip $u < n < 2u$ and has $D = 2^n - 3^u > 0$. The bridge, residue-profile, factorization, and plateau results are developed under the following standing conditions:

$$\boxed{z := n - u, \qquad 0 < z < u, \qquad \gcd(n, u) = \gcd(u, z) = 1, \qquad D := 2^n - 3^u > 0.} \tag{12}$$

Since a proper q-fold cyclic repetition would imply $q \mid n$ and $q \mid u$, coprimality prevents such a repetition; an integral realization therefore has least period n. The mechanical and one-swap mechanical exclusions stated in Section 1.3 extend beyond (12) to every parameter pair with $1 \le u < n$ and $D > 0$.

1.2 From a cyclic parity class to a sparse boundary

To motivate the new coordinates, choose an odd root and write a parity word cyclically as u odd symbols separated by zero gaps

$$\gamma = (\gamma_0, \dots, \gamma_{u-1}), \qquad \gamma_t \ge 0, \qquad \sum_{t=0}^{u-1} \gamma_t = z. \tag{13}$$

Call the order $t = 0, \dots, u-1$ in which the odd symbols are encountered from the chosen root the **odd-phase order**. Thus t is the odd-phase index, whereas the position of that phase in the full parity word will be denoted by j_t.

Among all distributions of these z zeros, the mechanical word distributes them as evenly as integer constraints permit. Its cumulative zero count follows the lower integer approximation

$$A(t) := \left\lfloor \frac{zt}{u} \right\rfloor, \qquad 0 \le t \le u. \tag{14}$$

With this zero-count convention, A is the lower mechanical path; the corresponding parity class is the associated upper-Christoffel class.

Balanced and mechanical parity patterns have appeared in several parts of Collatz theory. Halbeisen and Hungerbühler [3] studied their extremal role in rational cycles, while López and Stoll [4] examined mechanical and Sturmian parity sequences through the 2-adic Collatz conjugacy. More recently, Knight [5] identifies the coprime high-cycle parity vector with the relevant upper-Christoffel word and proves that the associated rational high cycle cannot consist entirely of integers. Fernández and Ibáñez state in a 2026 preprint [6, Theorem 7.3] that, at fixed length and weight, the Christoffel class uniquely maximizes, up to rotation, their functional $C_{\min}$.

Here the mechanical parity word has a specific role. Such a word is not proposed as an approximation to a hypothetical cycle, nor does the argument depend on its extremality. It serves instead as a coordinate origin for every cyclic parity class with the same length and weight.

A cyclic parity class has no distinguished first odd phase, and comparison with A depends on where the class is rooted. Under the coprimality hypothesis, Section 2.2 proves that there is a unique odd root for which the cumulative zero path

$$P(t) := \sum_{s<t} \gamma_s \tag{15}$$

never rises above $A(t)$. At this root, the vertical separation

$$L_t := A(t) - P(t) \tag{16}$$

is the **canonical bridge**.

The bridge begins and ends at zero and is nonnegative at every intermediate odd phase. Its value $L_t = A(t) - P(t)$ is the **bridge height** at odd phase t: the vertical distance from the cumulative path P to the mechanical path A. A positive height means that, up to that point, the given parity class has placed fewer even phases than the mechanical word and must repay that displacement later. Together with the fixed mechanical increments $a_t := A(t+1) - A(t)$, the bridge increments recover the zero gaps:

$$\gamma_t = a_t - (L_{t+1} - L_t). \tag{17}$$

Thus the bridge determines the canonically rooted word. Section 2.2 gives the precise legality conditions and inverse reconstruction. The zero bridge is exactly the balanced mechanical–Christoffel class.

The following nonintegral example is used only to illustrate the coordinates, including Figures 1 and 2. For

$$(n, u, z) = (8, 5, 3), \tag{18}$$

the mechanical path has increments $a = (0, 1, 0, 1, 1)$, giving the rooted mechanical reference word

$$w_{\text{mech}} = 11011010. \tag{19}$$

For comparison, take the canonically rooted word

$$w = 11101100. \tag{20}$$

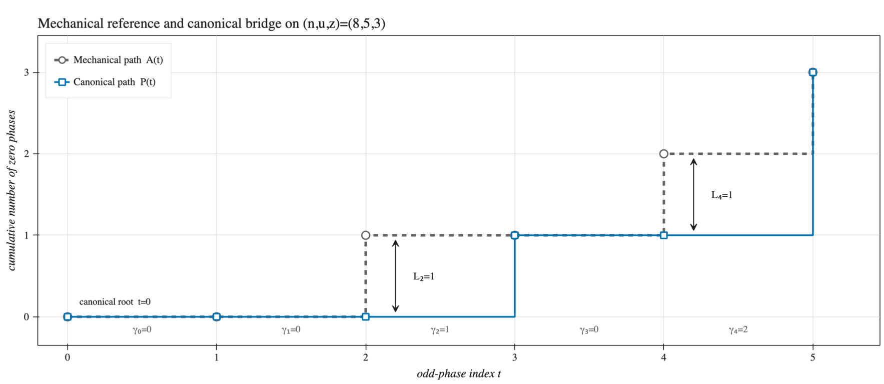

Figure 1: *Canonical bridge for the nonintegral example* $w = 11101100$. *The cumulative path P lies below the mechanical path A, and their vertical separation is* $L_t = A(t) - P(t)$.

Reading cyclically from one 1 to the next gives

$$\begin{aligned} \gamma &= (0,0,1,0,2), \qquad & P &= (0,0,0,1,1,3), \\ A &= (0,0,1,1,2,3), \qquad & L &= A - P = (0,0,1,0,1,0). \end{aligned} \tag{21}$$

For this word,

$$D = 13, \qquad N(w) = 251 \not\equiv 0 \pmod{13}. \tag{22}$$

The bridge now describes the parity class geometrically, but integral realizability is still tested through the mixed-base numerator $N(w)$. To connect these descriptions, the next step is to determine how each bridge height changes the numerator summand contributed by its odd phase.

Let j_t be the position of the tth odd phase in the full parity word, with t counted from the canonical root. Its numerator contribution is the mixed-base monomial $2^{j_t}3^{u-1-t}$. The mechanical word would place this phase at $t + A(t)$, so $j_t = t + A(t) - L_t$: a positive bridge height moves the odd symbol L_t positions earlier and changes its power of 2. This effect is obscured because the power of 3 is controlled by the different index t.

To compare these two exponent contributions, we seek a modular unit that turns each product 2^a3^b into a single power. The denominator gives the necessary compatibility:

$$D = 2^n - 3^u, \qquad 2^n \equiv 3^u \pmod{D}. \tag{23}$$

Since $\gcd(n,u) = 1$, we may use Bézout's identity to choose s, k with $su - kn = 1$ and define $\xi := 2^s3^{-k}$ in $(\mathbb{Z}/D\mathbb{Z})^\times$. A direct calculation using this congruence gives

$$\xi^u = 2, \qquad \xi^n = 3. \tag{24}$$

Thus, modulo D, every product 2^a3^b becomes the single power ξ^{ua+nb}. The construction is independent of the Bézout pair and uniquely determines ξ; these facts are proved in §3.1. Applied to the tth numerator term, it gives

$$2^{j_t}3^{u-1-t} \equiv \xi^{n(u-1)}\xi^{-(nt-uj_t)} \pmod{D}. \tag{25}$$

The common factor $\xi^{n(u-1)}$ is independent of t. The remaining exponent therefore provides the coordinate relevant to the bridge. Substituting $j_t = t + A(t) - L_t$ gives its quotient–remainder decomposition

$$nt - uj_t = r_t + uL_t, \qquad r_t \equiv (n-u)t \pmod{u}, \qquad 0 \le r_t < u. \tag{26}$$

This is an affine separation at each phase. The residue r_t locates the term within a block of u consecutive powers of ξ, while the height L_t counts whole blocks. Since $\xi^u = 2$, those whole blocks become dyadic weights; after normalization by the maximum height M, the weight is 2^{M-L_t}.

This separation motivates residue order. Since $\gcd(u, n-u) = 1$, the residues r_t run through $0, \dots, u-1$ exactly once. If $t(r)$ is the phase carrying residue r, define the residue-ordered bridge and its dyadic normalization by

$$H_r := L_{t(r)} \quad (0 \le r < u), \qquad M := \max_{0 \le r < u} H_r, \qquad d_r := 2^{M-H_r}. \tag{27}$$

This places each bridge height at the arithmetic location selected by its numerator term. Thus $H = (H_0, \dots, H_{u-1})$ is the canonical bridge written in residue order rather than odd-phase order. The coefficients d_r are precisely the height-dependent dyadic terms produced by the normalization above. Call $d = (d_0, \dots, d_{u-1})$ the **positive dyadic profile** of L in residue order. Its **profile polynomial** is

$$V_L(X) := \sum_{r=0}^{u-1} d_r X^{u-1-r}. \tag{28}$$

Because $h \mapsto 2^{M-h}$ is injective, the constant intervals of H coincide exactly with those of the coefficient sequence d.

The geometric information in V_L is therefore concentrated at the transitions between its constant intervals. In the quotient ring

$$\mathscr{A}_u := \mathbb{Z}[X]/(X^u - 2), \tag{29}$$

the associated **boundary polynomial** is

$$Q_L := [(X-1)V_L]_{X^u=2}. \tag{30}$$

Multiplication by $X - 1$ is invertible in $\mathscr{A}_u$, since

$$(X-1)(1 + X + \cdots + X^{u-1}) = X^u - 1 = 1 \qquad \text{in } \mathscr{A}_u. \tag{31}$$

Thus the passage from V_L to Q_L loses no information. The boundary determines the profile, the profile determines the bridge, and the bridge determines the original cyclic parity class.

Let $K(L)$ denote the number of maximal constant intervals in the linear residue-ordered profile, with the cut fixed by the canonical root. The support of Q_L has one occupied index for each maximal constant interval: one at each interior transition and one additional index arising from the reduction $X^u = 2$. Thus

$$\#\operatorname{supp} Q_L = K(L). \tag{32}$$

Residue order need not preserve adjacency in odd-phase order. Accordingly, $K(L)$ does not measure the visual roughness of the original bridge $t \mapsto L_t$; it measures how often the dyadic height changes in the arithmetic order imposed by the numerator.

The point of the boundary coordinate is arithmetic rather than merely descriptive. Theorem 1.1 shows that L is integrally realizable exactly when there is a reduced polynomial $C_L \in \mathbb{Z}[X]_{<u}$ such that

$$Q_L = (2X^z - 3)C_L \qquad \text{in } \mathscr{A}_u. \tag{33}$$

For such a realization, every one of the u coefficients of C_L in the basis $1, X, \ldots, X^{u-1}$ is nonzero. Thus the $K(L)$ occupied coefficients of Q_L are sparse forcing sites from which a full-support arithmetic quotient must be generated.

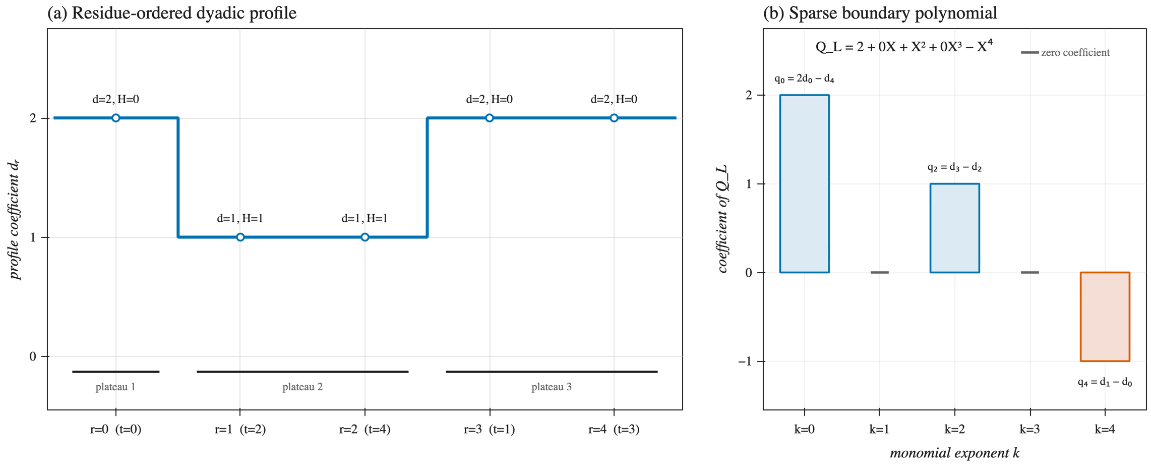

Figure 2: *Residue reordering and boundary formation for the nonintegral example $w = 11101100$. The profile $d = (2, 1, 1, 2, 2)$ has three maximal constant intervals, while $Q_L = 2 + X^2 - X^4$ has exactly three occupied coefficients.*

The construction may be summarized as

$$\text{cyclic parity class} \longrightarrow \text{canonical bridge} \longrightarrow \text{residue profile} \longrightarrow \text{sparse boundary}. \tag{34}$$

The first passage resolves cyclic rooting, the second separates mechanical displacement from arithmetic position, and the third concentrates the resulting geometry into a sparse coordinate.

1.3 Algebraic representation and main results

The boundary coordinate is constructed to preserve integral realizability, and the connection is exact. Combining the common modular scale and exponent decomposition developed above with the dyadic normalization defining V_L gives, for the canonically rooted word $w(L)$,

$$V_L(\xi) = 2^M \xi^{u-1} 3^{1-u} N(w(L)) \quad \text{in } \mathbb{Z}/D\mathbb{Z}. \tag{35}$$

The prefactor is a unit modulo D. Hence the original divisibility condition is equivalent to $V_L(\xi) = 0$. Since $Q_L = (X-1)V_L$ in $\mathcal{A}_u$ and $X - 1$ is a unit in that ring, its value $\xi - 1$ is a unit modulo D. The same condition is therefore equivalent to

$$Q_L(\xi) = 0. \tag{36}$$

For the $(8, 5, 3)$ example, one may take $\xi = 6$ modulo 13. The boundary coordinate then gives the same obstruction directly:

$$Q_L(\xi) = Q_L(6) \equiv 3 \not\equiv 0 \pmod{13}. \tag{37}$$

Thus evaluation at ξ carries the sparse boundary back to the original integral-realization problem. Section 4 identifies the resulting evaluation homomorphism

$$\mathrm{ev}_\xi : \mathcal{A}_u \longrightarrow \mathbb{Z}/D\mathbb{Z}, \tag{38}$$

and proves that its kernel is the principal ideal generated by

$$\Psi_{n,u}(X) := 2X^{n-u} - 3. \tag{39}$$

Modular vanishing is therefore equivalent to exact factorization by $\Psi_{n,u}$ in $\mathcal{A}_u$. Write

$$\mathbb{Z}[X]_{<u} := \{F \in \mathbb{Z}[X] : \deg F < u\}. \tag{40}$$

Every class in $\mathcal{A}_u$ has a unique representative in $\mathbb{Z}[X]_{<u}$. The quotient polynomials below always mean these reduced representatives, and the displayed factorization equalities are equalities in $\mathcal{A}_u$.

Theorem 1.1 (Boundary representation, realizability, and orbit recovery).

Assume the standing conditions (12), *let* $[w]_{\mathrm{cyc}}$ *be a cyclic parity class, let* L *be its canonical bridge, and let* $w(L)$ *be its canonically rooted representative. Then:*

(i) Canonical rooting gives a bijection between cyclic parity classes and canonical bridges, while the reduced boundary representative satisfies

$$[w]_{\mathrm{cyc}} \longleftrightarrow L, \qquad L \mapsto Q_L \text{ is injective}, \qquad \#\operatorname{supp} Q_L = K(L). \tag{41}$$

(ii) The following conditions are equivalent:

$$\begin{aligned} L \text{ is integrally realizable} \iff D \mid N(w(L)) \iff Q_L(\xi) = 0 \\ \iff Q_L \in \Psi_{n,u}\mathcal{A}_u. \end{aligned} \tag{42}$$

(iii) When these conditions hold, there are unique $Y_L, C_L \in \mathbb{Z}[X]_{<u}$ *such that*

$$V_L = \Psi_{n,u} Y_L, \qquad Q_L = \Psi_{n,u} C_L \qquad \text{in } \mathcal{A}_u. \tag{43}$$

Writing $Y_L = \sum_{e=0}^{u-1} b_e X^e$ *and* $e_t := u - 1 - r_t$, *its coefficients recover the positive odd orbit through*

$$x_t := \frac{[X^{u-1-r_t}]Y_L}{2^{M-L_t}} = \frac{b_{e_t}}{2^{M-L_t}} \in \mathbb{Z}_{>0} \cap (2\mathbb{Z}+1), \tag{44}$$

and the boundary quotient satisfies

$$C_L = [(X-1)Y_L]_{X^u=2}, \qquad \#\operatorname{supp} C_L = u. \tag{45}$$

The sparse boundary and full-support quotient in Theorem 1.1 drive the second main result concerning the number of residue-ordered plateaux.

For an integrally realizable bridge whose realization has least period n, the coefficient sequence of C_L has no zero entries. Each occupied coefficient of Q_L is a site at which the quotient recurrence may be inhomogeneous. With only $K(L)$ such sites, some cyclic gap has length at least $\lceil u/K(L) \rceil - 1$; across that gap the homogeneous recurrence forces a power of 3 into a nonzero quotient coefficient. Bridge geometry, meanwhile, bounds $\|Q_L\|_1$ in terms of $K(L)$, and Laurent's estimate prevents the normalized denominator from absorbing the resulting exponential discrepancy.

Balancing these two estimates, together with Laurent's two-logarithm estimate [7] applied to $n \log 2 - u \log 3$, gives the second main result.

Theorem 1.2 (Quantitative plateau complexity).

Let L be the canonical bridge of a nontrivial positive Collatz cycle with least period n and u odd members. If

$$\gcd(n, u) = 1, \tag{46}$$

then there exists an absolute constant $C > 0$ *such that, uniformly over all such cycles,*

$$K(L) \geq \sqrt{\log_2 3}\,\sqrt{u} - C(1 + \log u)^2. \tag{47}$$

Theorem 1.2 is therefore a deterministic complexity obstruction: too few geometric boundary sites force more arithmetic growth than the bridge can supply.

Existing global results also give a numerical lower bound on the parameters of any hypothetical nontrivial positive cycle. Hercher's Corollary 29 [8, see also 9], combined with Bařina's verification of convergence below 2^{71} [10], implies that its number u of odd members must satisfy $u > 1.375 \times 10^{11}$. Theorem 1.2 imposes a different structural requirement: the boundary complexity in residue order must itself grow at least on the square-root scale.

This invariant records different information from earlier cycle-complexity measures. Brox's descent number [11] and the m-cycle counts used by Simons and collaborators [12–14] measure block structure in the original parity-word or orbit order. By contrast, $K(L)$ counts constant intervals only after canonical rooting and reindexing the odd phases in residue order. No direct formula or comparison inequality connecting these quantities is established.

The two classes nearest the mechanical origin give unconditional exclusions. In Knight's terminology, the coprime upper-Christoffel word is the high-cycle parity vector for the fixed length and weight. Knight proves that, apart from the trivial Collatz cycle, the associated high cycle cannot consist entirely of positive integers [5]. In the present coordinates this class is represented by the zero bridge

$$L = 0 \tag{48}$$

which is exactly the balanced mechanical–Christoffel class. Its boundary is

$$Q_0 = 1, \tag{49}$$

which cannot contain the nonunit factor $\Psi_{n,u}$. This gives a boundary-theoretic recovery of Knight's high-cycle exclusion.

The immediately adjacent class has an equally direct description in parity-word coordinates. A **one-swap mechanical word** is any rooted word obtained from the rooted mechanical word by one internal adjacent swap

$$01 \mapsto 10. \tag{50}$$

In bridge coordinates, these words are exactly the unit spikes

$$L_t = \mathbf{1}_{\{t=p\}}, \qquad 1 \leq p < u. \tag{51}$$

called singleton bridges because the support of L is a singleton. The reconstruction identity $\gamma_t - a_t = L_t - L_{t+1}$ shows directly that the spike transfers one zero from mechanical gap $p - 1$ to the adjacent gap p, or equivalently moves the pth noninitial 1 one position earlier across the preceding 0.

Section 6 proves the resulting one-swap exclusion without a coprimality assumption: for every parameter pair satisfying

$$1 \leq u < n, \qquad 2^n - 3^u > 0, \tag{52}$$

neither the rooted mechanical word nor any of its one-swap mechanical words can encode a nontrivial positive integer Collatz cycle.

In the coprime branch, a singleton bridge has a three-term boundary whose support gap is incompatible with factorization once Eliahou's cycle-length bound [15] is applied. In the noncoprime branch, the denominator has a geometric cofactor that the single moved numerator term cannot absorb. These two branches exhaust the positive-denominator domain.

This exclusion concerns every internal adjacent swap, but not an arbitrary relocation of one zero. Moving a zero from a gap i to a later nonadjacent gap j, with $0 \leq i < j < u$ and $j \geq i + 2$, produces instead the height-one interval bridge $L_t = \mathbf{1}_{\{i<t\leq j\}}$ after a suitable rooting. That larger family is not excluded here. More generally, the full boundary and plateau theory remains coprime: when $\gcd(n, u) > 1$, the residue map is no longer a permutation, so a general extension must retain the multiple contributions at each reduced residue and control their possible cancellation.

1.4 Paper organization and theorem dependencies

The argument has one main proof spine and one special-family branch:

Part	Role in the argument	Principal result
Section 2	canonical rooting and bridge reconstruction	cyclic parity class $\longleftrightarrow L$
Section 3	residue order, dyadic profile, and boundary compression	$L \mapsto Q_L$ and $\#\operatorname{supp} Q_L = K(L)$
Section 4	principal kernel, exact factorization, and orbit recovery	$Q_L = (2X^z - 3)C_L$ with full quotient support
Section 5	boundary-gap growth and dyadic mass bounds	Theorem 1.2
Section 6	mechanical and one-swap mechanical words	full positive-denominator exclusion
Section 7	interpretation, losses, and extensions	open structural comparisons
Appendix A	explicit logarithmic estimates	analytic inputs to Sections 5 and 6

The main theorem spine is Section 2 $\longrightarrow$ Section 3 $\longrightarrow$ Section 4 $\longrightarrow$ Section 5.

Section 6 uses the coprime boundary theory on one branch and a separate geometric-cofactor argument on the noncoprime branch. Section 7 then compares the resulting invariant with native-order cycle complexity and records the remaining extension problems.

2 Canonical rooting and bridge reconstruction

2.1 Rational realization and the cycle strip

§1.1 records the affine identity $2^n T_w(x) = 3^u x + N(w)$ and the divisibility criterion $D \mid N(w)$ for integral realizability. The following proposition proves the identity and the parity realization that makes this the correct framework. Proposition 2.2 then establishes the strip in which nontrivial positive cycles must lie.

Proposition 2.1 (Prescribed-parity realization).

Let $w = (w_0, \dots, w_{n-1}) \in \{0,1\}^n$ have weight u, with $D := 2^n - 3^u$. For every $x \in \mathbb{Z}_{(2)}$,

$$2^n T_w(x) = 3^u x + N(w). \tag{53}$$

When $D \neq 0$, the affine map T_w has the unique fixed point $x(w) = N(w)/D$ in $\mathbb{Z}_{(2)}$, and its parity word under the parity-selected extension of T is exactly w. When $D > 0$ and $u > 0$, the word is integrally realizable if and only if $D \mid N(w)$.

Proof. If $\varepsilon \in \{0,1\}$ is appended to a word of length n, then

$$N(w\varepsilon) = 3^\varepsilon N(w) + \varepsilon 2^n. \tag{54}$$

Together with $2T_\varepsilon(x) = 3^\varepsilon x + \varepsilon$, this recurrence proves by induction that

$$2^n T_w(x) = 3^u x + N(w). \tag{55}$$

The fixed-point equation is $(2^n - 3^u)x = N(w)$, proving the fixed-point formula. Since D is odd, $x(w) \in \mathbb{Z}_{(2)}$.

Let σ denote left cyclic shift, with $x_0 := x(w)$, and for $1 \le i < n$ define

$$x_i := T_{w_{i-1}} \circ \cdots \circ T_{w_0}(x(w)). \tag{56}$$

The point x_i is fixed by $T_{\sigma^i w}$, so uniqueness of the affine fixed point gives

$$x_i = x(\sigma^i w) = \frac{N(\sigma^i w)}{D}. \tag{57}$$

For every binary word v, one has $N(v) \equiv v_0 \pmod 2$: the term at $j = 0$ has parity v_0, and every later term is even. Because D is odd, reducing $N(\sigma^i w)/D$ cancels only odd factors. The reduced numerator of x_i therefore has parity w_i. Thus the parity-selected map follows the prescribed branches, and the parity word of $x(w)$ is exactly w.

The identity $x_i = x(\sigma^i w)$ also shows that cyclic rotations merely select different starting points on the same rational orbit and preserve integral realizability.

When $D > 0$ and $u > 0$, both D and $N(w)$ are positive. Thus $x(w)$ is a positive integer exactly when $D \mid N(w)$. In that case all iterates in the resulting periodic orbit are integers. ☐

Proposition 2.2 (Cycle-relevant strip).

Let a nontrivial positive Collatz cycle have least period n and exactly u odd members. Then

$$u < n < 2u, \qquad 0 < z = n - u < u, \qquad D = 2^n - 3^u > 0. \tag{58}$$

Proof. The cycle contains an odd member, since a positive all-even orbit strictly decreases. Rooting its parity word anywhere on the cycle, Proposition 2.1 gives

$$Dx = N(w) > 0, \tag{59}$$

so $D > 0$. Hence $2^n > 3^u$ and therefore

$$n > u \log_2 3 > u. \tag{60}$$

List the odd members cyclically as $y_0, \dots, y_{u-1}$, with indices taken modulo u, and let γ_t be the number of even members strictly between y_t and y_{t+1}. A nontrivial cycle cannot contain 1, because the orbit through 1 is the cycle $(1, 2)$. Thus $y_t > 1$, and

$$2^{\gamma_t+1} y_{t+1} = 3y_t + 1 < 4y_t. \tag{61}$$

Multiplying over t mod u and cancelling $\prod_t y_t$ gives

$$2^{\sum_t(\gamma_t+1)} < 4^u. \tag{62}$$

The exponent on the left is the total period n, so $n < 2u$. The assertions for $z = n - u$ follow. □

Until explicitly stated otherwise, fix integers n, u satisfying

$$\gcd(n, u) = 1, \qquad z = n - u, \qquad 0 < z < u, \qquad D = 2^n - 3^u > 0. \tag{63}$$

Then $\gcd(u, z) = 1$.

The following elementary consequence of the standing assumptions also holds: $D > 1$. Indeed, suppose $D = 1$, so $2^n = 3^u + 1$. If u were even, then $u \geq 2$ and $3^u + 1 \equiv 2 \pmod 8$, which is impossible because $n > u$ gives $n \geq 3$. Thus u is odd. Then $3^u + 1 \equiv 4 \pmod 8$, so $n = 2$ and hence $u = 1$, contrary to $0 < n - u < u$.

2.2 Mechanical gaps and canonical rotation

Root a length-n parity word at one of its 1-symbols. Reading cyclically from that root, let

$$\gamma_t \in \mathbb{Z}_{\geq 0} \qquad (0 \leq t < u) \tag{64}$$

be the number of 0-symbols following the tth 1-symbol and preceding the next one. Thus the word has the cyclic form

$$1\,0^{\gamma_0}\,1\,0^{\gamma_1} \cdots 1\,0^{\gamma_{u-1}}, \tag{65}$$

and

$$\sum_{t=0}^{u-1} \gamma_t = z. \tag{66}$$

Call the order $t = 0, \dots, u-1$ in which the odd symbols are encountered from the chosen root the **odd-phase order**. The index t numbers an odd phase; it is distinct from the full-word position j_t introduced below.

For the chosen odd root, define

$$P(t) := \sum_{i=0}^{t-1} \gamma_i \qquad (0 \leq t \leq u). \tag{67}$$

Then the position of the tth 1-symbol is

$$j_t = t + P(t). \tag{68}$$

The balanced reference path and its increments are

$$A(t) := \left\lfloor \frac{tz}{u} \right\rfloor \quad (0 \le t \le u), \qquad a_t := A(t+1) - A(t) = \left\lfloor \frac{(t+1)z}{u} \right\rfloor - \left\lfloor \frac{tz}{u} \right\rfloor. \tag{69}$$

Because $0 < z < u$, each a_t belongs to $\{0, 1\}$, and

$$\sum_{t=0}^{u-1} a_t = z. \tag{70}$$

The sequence $a = (a_0, \dots, a_{u-1})$ distributes the z zero gaps as evenly as possible: at every prefix length t, its cumulative zero count $A(t)$ satisfies

$$0 \le \frac{tz}{u} - A(t) < 1. \tag{71}$$

The odd-rooted parity word with zero-gap sequence a is the mechanical–Christoffel representative.

The zero-gap data are cyclic, whereas their cumulative path P, and hence its displacement from A, depends on the chosen odd root. A useful coordinate system therefore requires a canonical rotation. The next theorem shows that coprimality provides one: exactly one rotation has cumulative zero count everywhere at or below the mechanical path. It then records the displacement from that path and proves that no information is lost.

Theorem 2.3 (Canonical bridge theorem).

Assume only

$$0 < z < u, \qquad \gcd(u, z) = 1, \tag{72}$$

and let

$$\boldsymbol{\gamma} = (\gamma_t)_{t \in \mathbb{Z}/u\mathbb{Z}} \tag{73}$$

be a cyclic nonnegative zero-gap sequence with total sum z.

(i) Among its u odd-rooted rotations, there is a unique rotation for which

$$P(t) \le A(t) \qquad (0 \le t \le u). \tag{74}$$

(ii) For this rotation, the bridge

$$L_t := A(t) - P(t) \qquad (0 \le t \le u) \tag{75}$$

*satisfies the **bridge legality conditions***

$$L_0 = L_u = 0, \qquad L_t \in \mathbb{Z}_{\ge 0}, \qquad L_{t+1} \le L_t + a_t \qquad (0 \le t < u). \tag{76}$$

In particular,

$$0 \le L_t \le A(t) \qquad (0 \le t \le u). \tag{77}$$

(iii) Conversely, every integer sequence $L = (L_t)_{t=0}^{u}$ satisfying the bridge legality conditions determines a unique canonical zero-gap sequence by

$$\gamma_t = a_t + L_t - L_{t+1}. \tag{78}$$

Proof. Existence and uniqueness of the root. Choose an arbitrary odd root and extend its cumulative zero count by

$$P(t+u) = P(t) + z. \tag{79}$$

Define

$$\Delta_t := uP(t) - tz. \tag{80}$$

The sequence Δ_t is u-periodic. Moreover, the values $\Delta_0, \dots, \Delta_{u-1}$ are pairwise distinct. Indeed, if $0 \le i < j < u$ and $\Delta_i = \Delta_j$, then

$$u\big(P(j) - P(i)\big) = (j-i)z. \tag{81}$$

Since $\gcd(u, z) = 1$, this would imply $u \mid j - i$, which is impossible.

Let s be the unique index at which Δ_t is maximal, and rotate the zero-gap sequence so that its new root is s. Its cumulative count is

$$P_s(t) := P(s+t) - P(s), \tag{82}$$

and therefore

$$uP_s(t) - tz = \Delta_{s+t} - \Delta_s < 0 \qquad (1 \le t < u). \tag{83}$$

Because tz/u is nonintegral for $1 \le t < u$, this strict inequality is equivalent to

$$P_s(t) \le \left\lfloor \frac{tz}{u} \right\rfloor = A(t). \tag{84}$$

Equality holds at $t = 0$ and $t = u$. Thus the maximizing root has the required property.

Conversely, if a root satisfies $P_s(t) \le A(t)$ for all t, then for $1 \le t < u$ one has $\Delta_{s+t} < \Delta_s$. Hence s is the unique maximizer, proving uniqueness of the root.

Bridge legality. For this root, $L_t = A(t) - P_s(t)$ is nonnegative and vanishes at $t = 0, u$. Since

$$L_{t+1} - L_t = a_t - \gamma_t, \tag{85}$$

one has $L_{t+1} \le L_t + a_t$. Since $L_0 = A(0) = 0$, this inequality also gives $L_t \le A(t)$ by induction.

Converse reconstruction. Suppose L satisfies the bridge legality conditions (76) and define

$$\gamma_t = a_t + L_t - L_{t+1}. \tag{86}$$

The bridge inequality gives $\gamma_t \geq 0$, while telescoping gives

$$\sum_{t=0}^{u-1} \gamma_t = \sum_{t=0}^{u-1} a_t + L_0 - L_u = z. \tag{87}$$

Its cumulative sum is

$$\sum_{i=0}^{t-1} \gamma_i = A(t) - L_t \leq A(t), \tag{88}$$

so the resulting zero-gap sequence is already in its canonical rotation. The two constructions are inverse. □

A sequence satisfying the bridge legality conditions is called a **canonical bridge** for (n, u). Since $P(t) = A(t) - L_t$, the height L_t is exactly the number of mechanical zeros not yet placed after t odd phases. The zero bridge gives the mechanical–Christoffel representative.

2.3 Reconstruction, realizability, and least period

Let L be a canonical bridge. Its zero gaps and odd positions are

$$\gamma_t = a_t + L_t - L_{t+1}, \qquad j_t = t + A(t) - L_t. \tag{89}$$

Since

$$A(t) - L_t = \sum_{i=0}^{t-1} \gamma_i, \tag{90}$$

one has

$$j_t = t + \sum_{i=0}^{t-1} \gamma_i. \tag{91}$$

Thus

$$0 = j_0 < j_1 < \cdots < j_{u-1} \leq n - 1, \tag{92}$$

and these are exactly the positions of the 1-symbols in the canonical odd-rooted parity word $w(L)$. The bridge therefore reconstructs both the zero gaps and the complete parity word. Define its numerator directly in bridge coordinates by

$$N(L) := N(w(L)) = \sum_{t=0}^{u-1} 2^{j_t} 3^{u-1-t}. \tag{93}$$

Corollary 2.4 (Bridge realizability criterion).

A canonical bridge L is integrally realizable if and only if

$$D \mid N(L). \tag{94}$$

Proof. This is Proposition 2.1 applied to the word whose odd positions are $j_0, \dots, j_{u-1}$. ☐

The bridge has thus retained the exact numerator and divisibility problem while removing the cyclic rooting ambiguity. For an integrally realizable bridge, the next proposition shows that coprimality removes the remaining ambiguity between the displayed period and the least period.

Proposition 2.5 (Least period under coprimality).

Suppose a length-n parity word with u ones is realized by a periodic orbit of least period d. Then

$$\frac{n}{d} \mid \gcd(n, u). \tag{95}$$

In particular, under the standing assumption $\gcd(n, u) = 1$*, every integrally realizable canonical bridge produces an orbit of least period n. Its u odd members are therefore pairwise distinct.*

Proof. Since $T^n(x) = x$ and d is the least positive return time, $d \mid n$; write $n = kd$. The length-n parity word consists of k repetitions of the least-period parity word. If one period contains u_0 ones, then $u = ku_0$. Thus $k = n/d$ divides $\gcd(n, u)$. Under the standing coprimality assumption, if $d < n$, then $k > 1$ divides $\gcd(n, u) = 1$, a contradiction. Hence $d = n$. ☐

3 Residue order and boundary compression

§1.1 introduced residue order as the arithmetic ordering that separates bridge height from numerator position. This section derives that coordinate from the canonical bridge formulas and uses it to place the cycle numerator on a common modular exponent scale.

Recall that $z = n - u$ and $j_t = t + A(t) - L_t$. Euclidean division of zt by u gives

$$zt = uA(t) + r_t, \qquad r_t := zt - uA(t), \qquad 0 \le r_t < u, \qquad r_t \equiv zt \pmod{u}. \tag{96}$$

Therefore

$$nt - uj_t = zt - uA(t) + uL_t = uL_t + r_t. \tag{97}$$

r_t is the least nonnegative residue of zt modulo u: L_t records the whole u-blocks in $nt - uj_t$, while r_t records the remaining arithmetic position.

Because $\gcd(u, z) = 1$, the map $t \mapsto r_t$ is a permutation of $\{0, \dots, u-1\}$. Let $t(r)$ denote its inverse. As defined in §1.1, increasing t is odd-phase order, whereas increasing r, or equivalently reading the phases as $t(0), \dots, t(u-1)$, is residue order. Passing between the two orders is therefore lossless.

For the example of §1.1 with $(n, u, z) = (8, 5, 3)$ and canonically rooted parity word $w = 11101100$, the two orders are related by

$$\begin{array}{r|ccccc} \text{odd-phase coordinate } t & 0 & 1 & 2 & 3 & 4 \\ \text{associated residue } r_t & 0 & 3 & 1 & 4 & 2 \\ \hline \text{residue coordinate } r & 0 & 1 & 2 & 3 & 4 \\ \text{odd phase at residue } r:\ t(r) & 0 & 2 & 4 & 1 & 3 \end{array} \tag{98}$$

This decomposition places the powers of 2 and 3 on a common modular exponent scale, converts the bridge heights into a positive dyadic profile, and then compresses that profile to its sparse discrete boundary.

3.1 The distinguished modular unit

The integer D is odd, and $D \equiv 2^n \not\equiv 0 \pmod 3$. Hence $\gcd(D, 6) = 1$, so both 2 and 3 are units in $\mathbb{Z}/D\mathbb{Z}$.

Lemma 3.1 (Distinguished modular unit).

There is a unique unit

$$\xi \in (\mathbb{Z}/D\mathbb{Z})^\times \tag{99}$$

such that

$$\xi^u = 2, \qquad \xi^n = 3. \tag{100}$$

Concretely, if $s, k \in \mathbb{Z}$ satisfy

$$su - kn = 1, \tag{101}$$

then

$$\xi = 2^s 3^{-k} \quad \textit{in } \mathbb{Z}/D\mathbb{Z}. \tag{102}$$

Proof. Since $\gcd(n, u) = 1$, such integers s, k exist. The congruence $2^n \equiv 3^u \pmod D$ gives

$$\xi^u = 2^{su} 3^{-ku} = 2\left(2^n 3^{-u}\right)^k = 2, \tag{103}$$

and

$$\xi^n = 2^{sn} 3^{-kn} = 3^{su-kn} = 3. \tag{104}$$

Here the second calculation again uses $2^n \equiv 3^u \pmod D$. If ζ is any other unit satisfying $\zeta^u = 2$ and $\zeta^n = 3$, then

$$\zeta = \zeta^{su-kn} = 2^s 3^{-k} = \xi. \tag{105}$$

Thus the unit is unique and independent of the chosen Bézout pair. □

All negative powers of 2, 3, and ξ below are taken in this unit group.

3.2 Common exponent scale in residue order

Applying Lemma 3.1 to (97) gives

$$\xi^{-uL_t - r_t} = 2^{-L_t} \xi^{-r_t} = \xi^{-nt+uj_t} = 3^{-t} 2^{j_t}. \tag{106}$$

Since $\xi^u = 2$, this separates the dyadic weighting contributed by L_t from the residue power ξ^{-r_t}. To organize the numerator by the consecutive residue powers $1, \xi^{-1}, \ldots, \xi^{-(u-1)}$, each bridge height must be indexed by the residue position of its phase. Accordingly, set

$$H_r := L_{t(r)} \qquad (0 \le r < u). \tag{107}$$

Because $r_0 = 0$ and $t(0) = 0$, one has $H_0 = L_0 = 0$.

3.3 Dyadic normalization and the profile polynomial

Let

$$M := \max_{0 \le r < u} H_r \tag{108}$$

be the maximum bridge height. To derive the positive dyadic weights associated with these heights, let $x_0 = N(L)/D \in \mathbb{Z}_{(2)}$ be the rational fixed point associated with $w(L)$. With the full-word positions j_t reconstructed in (89), define the remaining odd-phase values by

$$x_t := T^{j_t}(x_0) = T_{w_{j_t-1}} \circ \cdots \circ T_{w_0}(x_0) \qquad (1 \le t < u). \tag{109}$$

Set $j_u := n$ and $x_u := x_0$. Between x_t and x_{t+1} the orbit applies one odd branch followed by γ_t even branches, so

$$2^{\gamma_t+1} x_{t+1} = 3x_t + 1 \qquad (0 \le t < u). \tag{110}$$

Define the temporary dyadically normalized values

$$\hat{x}_t := 2^{M-L_t} x_t \qquad (0 \le t \le u). \tag{111}$$

Since $\gamma_t = a_t + L_t - L_{t+1}$, substitution gives the **dyadically normalized odd-phase recurrence**

$$2^{1+a_t}\hat{x}_{t+1} - 3\hat{x}_t = 2^{M-L_t} \qquad (0 \le t < u). \tag{112}$$

The right side depends only on the bridge height. Writing these source terms in the residue order introduced in §1.1 gives the positive dyadic profile

$$d_r := 2^{M-H_r} \qquad (0 \le r < u). \tag{113}$$

Thus $d_{r_t} = 2^{M-L_t}$ is precisely the source term in (112). At least one height equals M, so at least one d_r equals 1, while $d_0 = 2^M$.

Its profile polynomial is

$$V_L(X) := \sum_{r=0}^{u-1} d_r X^{u-1-r}. \tag{114}$$

The reverse exponent $u - 1 - r$ aligns evaluation at ξ with the factor 3^{u-1-t} in the cycle numerator.

Proposition 3.2 (Profile representation of the numerator).

In $\mathbb{Z}/D\mathbb{Z}$,

$$V_L(\xi) = 2^M \xi^{u-1} 3^{1-u} N(L). \tag{115}$$

In particular,

$$D \mid N(L) \quad \iff \quad V_L(\xi) = 0. \tag{116}$$

Proof. Using the numerator formula (93), reindex by $r = r_t$. With $H_{r_t} = L_t$ and (97),

$$\begin{aligned} V_L(\xi) &= \sum_{t=0}^{u-1} 2^{M-L_t} \xi^{u-1-r_t} \\ &= 2^M \xi^{u-1} \sum_{t=0}^{u-1} \xi^{-uL_t - r_t} \\ &= 2^M \xi^{u-1} \sum_{t=0}^{u-1} 3^{-t} 2^{j_t} \\ &= 2^M \xi^{u-1} 3^{1-u} \sum_{t=0}^{u-1} 2^{j_t} 3^{u-1-t} \\ &= 2^M \xi^{u-1} 3^{1-u} N(L). \end{aligned} \tag{117}$$

The prefactor is a product of units modulo D and is therefore itself a unit, proving the equivalence. $\square$

The profile polynomial carries the numerator's divisibility information, but every coefficient d_r is positive, so its support does not expose the bridge geometry. The discrete boundary exposes that geometry.

3.4 The boundary operator in $\mathscr{A}_u$

Because $\xi^u = 2$ modulo D, evaluation at ξ respects the polynomial relation $X^u = 2$. Accordingly, the discrete boundary is taken in the quotient ring

$$\mathscr{A}_u := \mathbb{Z}[X]/(X^u - 2). \tag{118}$$

Every class in $\mathscr{A}_u$ has a unique representative of degree less than u; write $[P]_{X^u=2}$ for this representative.

Multiplication by $X - 1$ is an integral automorphism of $\mathscr{A}_u$, because

$$(X-1)(1 + X + \cdots + X^{u-1}) = X^u - 1 = 1 \qquad \text{in } \mathscr{A}_u. \tag{119}$$

Multiplication by $X-1$ records the ordinary first differences at the interior residue positions. Reducing the remaining term $d_0 X^u$ by $X^u = 2$ produces the endpoint coefficient $2d_0 - d_{u-1}$. Therefore define the **boundary polynomial** of L by

$$Q_L(X) := [(X-1)V_L(X)]_{X^u=2}. \tag{120}$$

Expanding and reducing the single term of degree u gives

$$Q_L(X) = (2d_0 - d_{u-1}) + \sum_{r=1}^{u-1} (d_r - d_{r-1}) X^{u-r}. \tag{121}$$

Thus the operator in (120) produces ordinary first differences at the interior positions, while the endpoint coefficient retains the profile's overall level.

Let $K(L)$ denote the number of maximal intervals on which

$$H_0, H_1, \dots, H_{u-1} \tag{122}$$

is constant in the linear residue order $0, 1, \dots, u-1$. The endpoints are not identified: the endpoint coefficient of Q_L is counted separately from the interior transitions.

Theorem 3.3 (Canonical boundary coordinates).

Let L be any canonical bridge, including the zero bridge. Then:

1. *The boundary polynomial satisfies*

$$Q_L(1) = 2^M, \qquad \# \operatorname{supp} Q_L = K(L), \qquad \operatorname{cont}(Q_L) = 1, \tag{123}$$

   where $Q_L(1)$ denotes ordinary evaluation of the unique degree-$<$ u representative and cont *denotes the greatest common divisor of its coefficients.*
2. *The map*

$$L \longmapsto Q_L \tag{124}$$

   is injective. More precisely,

$$Q_L(X) = q_0 + \sum_{r=1}^{u-1} q_r X^{u-r}. \tag{125}$$

   Then the dyadic profile and bridge are recovered by

$$d_0 = Q_L(1), \qquad d_r = Q_L(1) + \sum_{i=1}^{r} q_i \quad (1 \le r < u), \tag{126}$$

$$M = \log_2 Q_L(1), \qquad H_r = M - \log_2 d_r, \qquad L_t = H_{r_t}. \tag{127}$$

Proof. For a general vector $v = (v_0, \dots, v_{u-1}) \in \mathbb{Z}^u$, define

$$R_v(X) := \sum_{r=0}^{u-1} v_r X^{u-1-r} \tag{128}$$

and define the boundary operator

$$\partial_u^{\mathrm{bd}} v := [(X-1)R_v]_{X^u=2}. \tag{129}$$

A direct expansion gives

$$\partial_u^{\mathrm{bd}} v = (2v_0 - v_{u-1}) + \sum_{r=1}^{u-1} (v_r - v_{r-1}) X^{u-r}. \tag{130}$$

If

$$\partial_u^{\mathrm{bd}} v = q_0 + \sum_{r=1}^{u-1} q_r X^{u-r}, \tag{131}$$

then $q_r = v_r - v_{r-1}$ for $r \ge 1$, and therefore

$$v_r = v_0 + \sum_{i=1}^{r} q_i. \tag{132}$$

Moreover,

$$\begin{aligned}(\partial_u^{\mathrm{bd}} v)(1) &= (2v_0 - v_{u-1}) + \sum_{r=1}^{u-1} (v_r - v_{r-1}) \\ &= v_0.\end{aligned} \tag{133}$$

Hence

$$v_0 = (\partial_u^{\mathrm{bd}} v)(1), \qquad v_r = (\partial_u^{\mathrm{bd}} v)(1) + \sum_{i=1}^{r} q_i. \tag{134}$$

The map $\partial_u^{\mathrm{bd}} : \mathbb{Z}^u \to \mathbb{Z}[X]_{<u}$ is therefore unimodular.

Applying (134) to $v = d$ gives $Q_L(1) = d_0 = 2^M$ and the stated reconstruction formulas. Since r_t is a permutation, the recovered profile H recovers every bridge value $L_t = H_{r_t}$. Thus $L \mapsto Q_L$ is injective.

For the support statement, the interior coefficient at X^{u-r} is nonzero exactly when $d_r \neq d_{r-1}$, or equivalently when $H_r \neq H_{r-1}$. The $K(L)$ linear plateaux contribute exactly $K(L) - 1$ nonzero interior transition coefficients. The endpoint coefficient is always nonzero, since

$$2d_0 - d_{u-1} = 2^{M+1} - d_{u-1} \geq 2^M > 0. \tag{135}$$

Hence $\# \operatorname{supp} Q_L = K(L)$.

Finally, some height equals M, so some $d_r = 1$. Thus the coefficient vector d has content 1. A unimodular transformation preserves the ideal generated by the coordinates, so $\operatorname{cont}(Q_L) = \operatorname{cont}(d) = 1$. ☐

The theorem gives the boundary two simultaneous interpretations. Geometrically, Q_L records the $K(L) - 1$ interior transitions of the residue-ordered profile together with one endpoint coefficient retaining its overall level. Algebraically, it is a primitive integral coordinate from which the entire bridge is reconstructed.

3.5 Modular criterion for integral realizability

The boundary operator preserves the divisibility information carried by the positive profile.

Theorem 3.4 (Modular boundary criterion).

For every canonical bridge L, the following conditions are equivalent:

$$\begin{aligned}&L \textit{ is integrally realizable}, \\ &D \mid N(L), \\ &V_L(\xi) = 0 \quad \textit{in } \mathbb{Z}/D\mathbb{Z}, \\ &Q_L(\xi) = 0 \quad \textit{in } \mathbb{Z}/D\mathbb{Z}.\end{aligned} \tag{136}$$

Proof. Corollary 2.4 gives the equivalence of the first two conditions, and Proposition 3.2 gives the equivalence of the second and third.

Because $\xi^u = 2$, evaluation at ξ respects reduction modulo $X^u - 2$. Hence

$$Q_L(\xi) = (\xi - 1)V_L(\xi). \tag{137}$$

The same geometric-sum identity used in $\mathscr{A}_u$ gives

$$(\xi - 1)(1 + \xi + \cdots + \xi^{u-1}) = \xi^u - 1 = 1 \qquad \text{in } \mathbb{Z}/D\mathbb{Z}. \tag{138}$$

Thus $\xi - 1$ is a unit, and multiplication by it preserves vanishing modulo D. □

The divisibility condition on the numerator has been replaced by modular vanishing of a primitive boundary whose support size is exactly the plateau count.

The distinguished modular unit also satisfies

$$2\xi^z = \xi^u \xi^z = \xi^n = 3. \tag{139}$$

Thus ξ is a common zero, modulo D, of $X^u - 2$ and

$$\Psi_{n,u}(X) := 2X^z - 3. \tag{140}$$

Since $\Psi_{n,u}$ and $X^u - 2$ are coprime over $\mathbb{Q}$, every profile V_L has a unique rational quotient by $\Psi_{n,u}$ in

$$\mathscr{A}_u^{\mathbb{Q}} := \mathbb{Q}[X]/(X^u - 2). \tag{141}$$

The $(8, 5, 3)$ example first makes this boundary construction concrete.

3.6 The $(8, 5, 3)$ example

$$(n, u, z) = (8, 5, 3), \qquad D = 2^8 - 3^5 = 13. \tag{142}$$

The mechanical path and increments are

$$A = (0, 0, 1, 1, 2, 3), \qquad a = (0, 1, 0, 1, 1). \tag{143}$$

Consider the bridge

$$L = (0, 0, 1, 0, 1, 0). \tag{144}$$

It satisfies the bridge conditions, and its associated coordinates are:

object	value
zero gaps $\gamma_t = a_t + L_t - L_{t+1}$	$(0, 0, 1, 0, 2)$
odd positions $j_t = t + A(t) - L_t$	$(0, 1, 2, 4, 5)$
residue labels in odd-phase order $(r_t)_{t=0}^4$	$(0, 3, 1, 4, 2)$
odd-phase indices in residue order $(t(r))_{r=0}^4$	$(0, 2, 4, 1, 3)$
bridge heights in residue order $H_r = L_{t(r)}$	$(0, 1, 1, 0, 0)$
dyadic profile $d_r = 2^{1-H_r}$	$(2, 1, 1, 2, 2)$

Thus $M = 1$ and

$$V_L(X) = 2X^4 + X^3 + X^2 + 2X + 2. \tag{145}$$

The height profile has three linear plateaux,

$$(0), \qquad (1,1), \qquad (0,0), \tag{146}$$

and its boundary is

$$Q_L(X) = [(X-1)V_L(X)]_{X^5=2} = 2 + X^2 - X^4. \tag{147}$$

Accordingly,

$$\#\operatorname{supp} Q_L = 3 = K(L), \qquad Q_L(1) = 2 = 2^M. \tag{148}$$

The cycle numerator is

$$N(L) = 3^4 + 2\cdot 3^3 + 2^2\cdot 3^2 + 2^4\cdot 3 + 2^5 = 251 \equiv 4 \pmod{13}. \tag{149}$$

Since $5u - 3n = 1$, one may take $(s,k) = (5,3)$ in Lemma 3.1, giving

$$\xi = 2^5 3^{-3} \equiv 6 \pmod{13}. \tag{150}$$

Evaluation of the boundary gives

$$Q_L(\xi) \equiv 2 + 6^2 - 6^4 \equiv 3 \not\equiv 0 \pmod{13}. \tag{151}$$

The bridge is therefore canonical but not integrally realizable. The example displays the complete passage

$$L = (L_t) \longrightarrow H = (H_r) \longrightarrow d = (d_r) \longrightarrow Q_L \longrightarrow Q_L(\xi). \tag{152}$$

That is, it passes from the canonical bridge in odd-phase order to the bridge heights in residue order, then to the dyadic profile, boundary polynomial, and modular realizability test.

4 Exact factorization and orbit recovery

Theorem 3.4 identified integral realization with modular boundary vanishing:

$$Q_L(\xi) = 0 \pmod{D}. \tag{153}$$

§3.5 observed that $\Psi_{n,u} = 2X^z - 3$ is invertible in $\mathscr{A}_u^{\mathbb{Q}} := \mathbb{Q}[X]/(X^u - 2)$, so the rational quotient

$$Y_L^{\mathbb{Q}} := \Psi_{n,u}^{-1} V_L \quad \text{in } \mathscr{A}_u^{\mathbb{Q}} \tag{154}$$

exists for every canonical bridge. The present section identifies the kernel of evaluation at ξ as the principal ideal $\Psi_{n,u}\mathscr{A}_u$ and proves that integral realization is precisely the condition that this quotient is integral. The resulting integral quotient recovers the odd orbit, while the boundary quotient has full support for least-period realizations.

Throughout this section,

$$\mathcal{A}_u := \mathbb{Z}[X]/(X^u - 2), \qquad \Psi_{n,u}(X) := 2X^z - 3, \tag{155}$$

and $\mathbb{Z}[X]_{<u}$ denotes the polynomials of degree less than u. Every class in $\mathcal{A}_u$ is represented by a unique element of $\mathbb{Z}[X]_{<u}$. Unless otherwise stated, polynomial identities in this section are identities in $\mathcal{A}_u$.

4.1 The evaluation kernel is principal

Define

$$W_{n,u}(X) := \sum_{k=0}^{u-1} 3^{u-1-k}(2X^z)^k. \tag{156}$$

The finite geometric identity gives

$$\Psi_{n,u}W_{n,u} = (2X^z)^u - 3^u = D \qquad \text{in } \mathcal{A}_u. \tag{157}$$

The next theorem shows that $\Psi_{n,u}$ generates the entire kernel of evaluation at ξ.

Theorem 4.1 (Principal evaluation kernel).

Evaluation at the distinguished modular unit induces a ring isomorphism

$$\mathcal{A}_u/\Psi_{n,u}\mathcal{A}_u \cong \mathbb{Z}/D\mathbb{Z}. \tag{158}$$

Equivalently, for every $P \in \mathbb{Z}[X]_{<u}$,

$$P(\xi) = 0 \pmod{D} \tag{159}$$

if and only if there is a unique $R \in \mathbb{Z}[X]_{<u}$ such that

$$P = \Psi_{n,u}R \qquad \textit{in } \mathcal{A}_u. \tag{160}$$

For this quotient,

$$DR = [PW_{n,u}]_{X^u=2}. \tag{161}$$

Proof. The strategy is to compare the index of the principal ideal $\Psi_{n,u}\mathcal{A}_u$ with the size of the target ring.

Multiplication by $\Psi_{n,u}$ is an endomorphism of the free abelian group $\mathcal{A}_u$. Its determinant is, up to sign, the resultant

$$\mathrm{Res}(X^u - 2, 2X^z - 3). \tag{162}$$

If $\alpha^u = 2$, then $\zeta = \alpha^z$ satisfies $\zeta^u = 2^z$. Both root sets contain u distinct elements. The map $\alpha \mapsto \alpha^z$ is injective between them: the ratio of any two roots of $X^u - 2$ is a uth root of unity, and exponentiation by z permutes the uth roots of unity because $\gcd(u, z) = 1$. It is therefore a bijection. Hence

$$
\begin{aligned}
|\mathrm{Res}(X^u - 2, 2X^z - 3)| &= \left| \prod_{\zeta^u = 2^z} (2\zeta - 3) \right| \\
&= |3^u - 2^{u+z}| \\
&= D.
\end{aligned} \tag{163}
$$

It follows that $\mathcal{A}_u / \Psi_{n,u} \mathcal{A}_u$ has exactly D elements. Evaluation at ξ is well defined on $\mathcal{A}_u$ because $\xi^u = 2$. Since $\Psi_{n,u}(\xi) = 2\xi^z - 3 = 0$ in $\mathbb{Z}/D\mathbb{Z}$, evaluation factors through

$$
\mathcal{A}_u / \Psi_{n,u} \mathcal{A}_u \longrightarrow \mathbb{Z}/D\mathbb{Z}. \tag{164}
$$

Every residue class modulo D is the image of a constant polynomial, so the induced map is surjective. Since its source and target both have D elements, it is an isomorphism. This identifies the evaluation kernel with $\Psi_{n,u}\mathcal{A}_u$.

The nonzero determinant computed in (163) also shows that multiplication by $\Psi_{n,u}$ is injective on $\mathcal{A}_u$, proving uniqueness of R. Finally, multiplying $P = \Psi_{n,u} R$ by $W_{n,u}$ and using (157) gives

$$
[PW_{n,u}]_{X^u = 2} = DR. \tag{165}
$$

□

Thus modular vanishing has become exact principal-ideal membership. Applying the theorem to the boundary polynomial gives a unique **boundary quotient** whenever the bridge is integrally realizable.

For an arbitrary canonical bridge, let $Y_L^{\mathbb{Q}}, C_L^{\mathbb{Q}} \in \mathbb{Q}[X]_{<u}$ be the unique representatives satisfying

$$
V_L = \Psi_{n,u} Y_L^{\mathbb{Q}}, \qquad Q_L = \Psi_{n,u} C_L^{\mathbb{Q}} \qquad \text{in } \mathcal{A}_u^{\mathbb{Q}}. \tag{166}
$$

Corollary 4.2 (Integral-lift criterion).

For every canonical bridge L, the following conditions are equivalent:

$$
L \text{ is integrally realizable} \quad \iff \quad Y_L^{\mathbb{Q}} \in \mathbb{Z}[X]_{<u} \quad \iff \quad C_L^{\mathbb{Q}} \in \mathbb{Z}[X]_{<u}. \tag{167}
$$

Proof. By Theorem 3.4, integral realizability is equivalent to either vanishing condition $V_L(\xi) = 0$ or $Q_L(\xi) = 0$. Theorem 4.1 identifies each condition with divisibility by $\Psi_{n,u}$ in $\mathcal{A}_u$, which is exactly integrality of the corresponding rational quotient. □

4.2 The profile and boundary quotients

Let L now be an integrally realizable canonical bridge. By Theorem 3.4,

$$
V_L(\xi) = Q_L(\xi) = 0 \pmod{D}. \tag{168}
$$

Theorem 4.1 therefore supplies unique polynomials

$$
Y_L, C_L \in \mathbb{Z}[X]_{<u} \tag{169}
$$

such that

$$V_L = \Psi_{n,u} Y_L, \qquad Q_L = \Psi_{n,u} C_L. \tag{170}$$

By Corollary 4.2, these are exactly the integral representatives of $Y_L^{\mathbb{Q}}$ and $C_L^{\mathbb{Q}}$. Call Y_L the **scaled-orbit quotient** and C_L the **boundary quotient**.

Theorem 4.3 (The quotient square).

For every integrally realizable canonical bridge:

1. *the quotient square commutes, equivalently*

$$C_L = [(X-1)Y_L]_{X^u=2}. \tag{171}$$

2. *every coefficient of Y_L is positive;*
3. *both Y_L and C_L are primitive.*

The square in part 1 is

$$\begin{array}{ccc} V_L & \xrightarrow{X-1} & Q_L \\ \Psi_{n,u}\uparrow & & \uparrow\Psi_{n,u} \\ Y_L & \xrightarrow{X-1} & C_L. \end{array} \tag{172}$$

Proof. Quotient square. By (120),

$$\Psi_{n,u} C_L = Q_L = (X-1)V_L = \Psi_{n,u}(X-1)Y_L. \tag{173}$$

Multiplication by $\Psi_{n,u}$ is injective by Theorem 4.1, so

$$C_L = (X-1)Y_L \qquad \text{in } \mathscr{A}_u. \tag{174}$$

Coefficientwise positivity. Theorem 4.1 gives

$$DY_L = [V_L W_{n,u}]_{X^u=2}. \tag{175}$$

Reduce $W_{n,u}$ first modulo $X^u - 2$. Since

$$X^{tz} = 2^{A(t)} X^{r_t}, \tag{176}$$

one obtains

$$[W_{n,u}]_{X^u=2} = \sum_{t=0}^{u-1} 2^{t+A(t)} 3^{u-1-t} X^{r_t}. \tag{177}$$

The residues r_t form a permutation, so this reduced polynomial has a strictly positive coefficient in every degree; in particular, its constant coefficient is positive. The same is true of V_L. For each output degree e, the coefficient of $[V_L W_{n,u}]_{X^u=2}$ receives the strictly positive product of the coefficient of X^e in V_L with the constant coefficient of the reduced $W_{n,u}$. Every other contribution is nonnegative, because reduction replaces each wrap by a positive power of 2. Hence every output coefficient is strictly positive. Since $D > 0$, every coefficient of Y_L is positive.

Primitivity. The profile V_L is primitive because one of its coefficients d_r equals 1. If an integer divided every coefficient of Y_L, it would divide every coefficient of $V_L = \Psi_{n,u} Y_L$; hence Y_L is primitive. The same argument, using the primitivity of Q_L from Theorem 3.3, shows that C_L is primitive. □

The four polynomials now have distinct roles:

polynomial	interpretation
V_L	positive dyadic height profile
Q_L	sparse boundary with $K(L) - 1$ interior transitions and one endpoint coefficient
Y_L	positive quotient containing the odd orbit
C_L	boundary quotient with full support for least-period realizations

The next theorem makes the orbit interpretation of Y_L exact.

4.3 The scaled-orbit quotient recovers the odd orbit

Write

$$Y_L(X) = \sum_{e=0}^{u-1} b_e X^e, \tag{178}$$

and retain the reverse residue indices

$$e_t := u - 1 - r_t \qquad (t \bmod u). \tag{179}$$

To write the quotient coefficients in odd-phase order, define

$$\eta_t := b_{e_t}. \tag{180}$$

For a nonzero integer m, let $v_2(m)$ denote its 2-adic valuation.

Theorem 4.4 (Orbit recovery).

The coefficients of Y_L satisfy

$$v_2(\eta_t) = M - L_t \qquad (t \bmod u). \tag{181}$$

Consequently, the quantities

$$\tilde{x}_t := \frac{\eta_t}{2^{M-L_t}} \tag{182}$$

are positive odd integers, and the sequence $(\tilde{x}_t)_{t \bmod u}$ obeys

$$2^{\gamma_t+1}\tilde{x}_{t+1} = 3\tilde{x}_t + 1. \tag{183}$$

These are precisely the odd members of the positive integral orbit determined by L, in odd-phase order. In particular, $\tilde{x}_t = x_t$ for the rational odd-phase values defined in §3.3.

Proof. We first extract the coefficient recurrence encoded by

$$V_L = (2X^z - 3)Y_L. \tag{184}$$

The residue relation

$$r_{t+1} = r_t + z - ua_t \tag{185}$$

is equivalent to

$$e_{t+1} + z = e_t + ua_t. \tag{186}$$

Therefore, in $\mathscr{A}_u$,

$$X^{e_{t+1}+z} = 2^{a_t} X^{e_t}. \tag{187}$$

Comparing the coefficient of X^{e_t} in (184) gives

$$2^{1+a_t}\eta_{t+1} - 3\eta_t = d_{r_t} = 2^{M-L_t}. \tag{188}$$

This is the integral coefficient form of the dyadically normalized odd-phase recurrence (112) derived from the rational orbit in §3.3.

Choose t_0 with $L_{t_0} = M$. At $t = t_0$, the right side equals 1, while the first term on the left is even. Hence η_{t_0} is odd and

$$v_2(\eta_{t_0}) = 0 = M - L_{t_0}. \tag{189}$$

We now propagate this valuation backward around the odd-phase cycle. Suppose

$$v_2(\eta_{t+1}) = M - L_{t+1}. \tag{190}$$

The bridge inequality $L_{t+1} \le L_t + a_t$ gives

$$\begin{aligned} v_2\big(2^{1+a_t}\eta_{t+1}\big) &= 1 + a_t + M - L_{t+1} \\ &\ge M - L_t + 1. \end{aligned} \tag{191}$$

This valuation is strictly larger than that of the right side 2^{M-L_t}. Equation (188) therefore forces

$$v_2(3\eta_t) = M - L_t, \tag{192}$$

and hence $v_2(\eta_t) = M - L_t$. Backward induction modulo u reaches every odd-phase index.

Theorem 4.3 gives $\eta_t > 0$, so $\tilde{x}_t$ is a positive odd integer. Substituting

$$\eta_t = 2^{M-L_t}\tilde{x}_t \tag{193}$$

into (188) and dividing by 2^{M-L_t} yields

$$2^{1+a_t+L_t-L_{t+1}}\tilde{x}_{t+1} - 3\tilde{x}_t = 1. \tag{194}$$

Because

$$\gamma_t = a_t + L_t - L_{t+1}, \tag{195}$$

this is exactly

$$2^{\gamma_t+1}\tilde{x}_{t+1} = 3\tilde{x}_t + 1. \tag{196}$$

Thus $\tilde{x}_{t+1}$ is obtained from $\tilde{x}_t$ by one odd Collatz step followed by precisely γ_t even steps. The resulting cyclic parity word is $w(L)$, whose rational periodic realization is unique by Proposition 2.1. Hence these $\tilde{x}_t$ equal the rational odd-phase values x_t defined in §3.3, now known to be positive odd integers, and $\eta_t = \hat{x}_t$. □

The theorem identifies the dyadic normalization in V_L: the bridge height L_t is exactly the amount of 2-adic scaling removed from the corresponding coefficient of Y_L to recover the odd orbit entry.

4.4 Least period and full quotient support

The boundary Q_L may be sparse, but its quotient C_L is not. Under the standing coprime hypotheses, Proposition 2.5 shows that every integrally realizable bridge has least period n; in particular, its odd entries $x_0, \dots, x_{u-1}$ are pairwise distinct.

Theorem 4.5 (Least period forces full support).

Let L be an integrally realizable canonical bridge. Then its realization has least period n, and every coefficient of C_L is nonzero. Equivalently,

$$\#\operatorname{supp} C_L = u. \tag{197}$$

Proof. The least-period assertion is Proposition 2.5. Reorder the coefficients of Y_L by the residue profile. For $0 \le r < u$, define

$$U_r := d_r x_{t(r)} = 2^{M-H_r} x_{t(r)}. \tag{198}$$

Theorem 4.4 gives

$$Y_L(X) = \sum_{r=0}^{u-1} U_r X^{u-1-r}. \tag{199}$$

Since $C_L = [(X-1)Y_L]_{X^u=2}$,

$$C_L(X) = (2U_0 - U_{u-1}) + \sum_{r=1}^{u-1} (U_r - U_{r-1}) X^{u-r}. \tag{200}$$

Suppose an interior coefficient vanished. Then $U_r = U_{r-1}$ for some $1 \le r < u$. Because the orbit entries are odd,

$$\nu_2(U_r) = M - H_r, \qquad \nu_2(U_{r-1}) = M - H_{r-1}. \tag{201}$$

Equality of the two integers first forces $H_r = H_{r-1}$ and hence $d_r = d_{r-1}$. Dividing by this common dyadic factor then gives

$$x_{t(r)} = x_{t(r-1)}. \tag{202}$$

The indices $t(r)$ and $t(r-1)$ are distinct, contradicting the pairwise distinctness of the odd orbit entries.

For the endpoint coefficient, $H_0 = 0$, so

$$v_2(2U_0) = M + 1. \tag{203}$$

On the other hand,

$$v_2(U_{u-1}) = M - H_{u-1} \leq M. \tag{204}$$

The two terms have different 2-adic valuations and therefore cannot cancel. Hence the endpoint coefficient is also nonzero. □

An integrally realizable bridge therefore produces a sparse–dense factorization:

$$Q_L = \Psi_{n,u} C_L, \qquad \#\operatorname{supp} Q_L = K(L), \qquad \#\operatorname{supp} C_L = u. \tag{205}$$

4.5 Proof of Theorem 1.1

Proof. Under the standing conditions, $0 < z < u$ and $\gcd(u,z) = \gcd(u,n) = 1$. Theorem 2.3 therefore gives the bijection between cyclic parity classes and canonical bridges asserted in part (i). Theorem 3.3 proves that $L \mapsto Q_L$ is injective and that $\#\operatorname{supp} Q_L = K(L)$.

For part (ii), Theorem 3.4 gives

$$L \text{ is integrally realizable} \iff D \mid N(w(L)) \iff Q_L(\xi) = 0. \tag{206}$$

Theorem 4.1 identifies the last condition with

$$Q_L \in \Psi_{n,u}\mathscr{A}_u, \tag{207}$$

which proves the stated chain of equivalences.

Suppose these conditions hold. Corollary 4.2 and Theorem 4.1 give the unique reduced integral quotients $Y_L, C_L \in \mathbb{Z}[X]_{<u}$ satisfying

$$V_L = \Psi_{n,u} Y_L, \qquad Q_L = \Psi_{n,u} C_L \qquad \text{in } \mathscr{A}_u. \tag{208}$$

Theorem 4.3 gives $C_L = [(X-1)Y_L]_{X^u=2}$ and coefficientwise positivity of Y_L. Theorem 4.4 then proves the coefficient formula recovering the positive odd orbit entries, including their oddness. Finally, Theorem 4.5 gives least period n and $\#\operatorname{supp} C_L = u$. These are exactly the assertions of part (iii). □

The next section quantifies the tension between these two support patterns. Long gaps in the sparse boundary force powers of 3 into the dense quotient, while the bridge geometry bounds the total mass of the boundary in terms of $K(L)$ alone.

5 Boundary gaps and plateau complexity

Section 4 produced the sparse–dense factorization $Q_L = \Psi_{n,u} C_L$, in which the boundary Q_L carries exactly $K(L)$ nonzero coefficients while the quotient C_L has full support. This section bounds the two structures and combines the bounds to prove the finite plateau theorem.

Let L be an integrally realizable canonical bridge, and set

$$Q := Q_L, \qquad C := C_L, \qquad K := K(L), \qquad M := \max_{0 \le r < u} H_r, \qquad \delta := \frac{D}{2^n}. \tag{209}$$

Under the standing conditions, Corollary 2.4 and Proposition 2.5 identify such an L with the canonical bridge of a positive integer cycle of least period n.

Such a bridge is necessarily nonzero. Indeed, for $L = 0$ one has $M = 0$, $d_r = 1$ for every r, and therefore

$$V_0 = 1 + X + \cdots + X^{u-1}, \qquad Q_0 = [(X-1)V_0]_{X^u = 2} = 1. \tag{210}$$

As noted in §2.1, the standing assumptions give $D > 1$, so $Q_0(\xi) = 1 \neq 0$ in $\mathbb{Z}/D\mathbb{Z}$, contrary to Theorem 3.4. Thus every integrally realizable bridge is nonzero and $M \ge 1$.

The argument first bounds C from below using boundary gaps, then bounds Q from above using the bridge geometry, and finally combines the two estimates.

For a polynomial $P = \sum_{e=0}^{u-1} p_e X^e$ and $1 \le p < \infty$, use

$$\|P\|_p := \left(\sum_{e=0}^{u-1} |p_e|^p \right)^{1/p}, \qquad \|P\|_\infty := \max_e |p_e|. \tag{211}$$

The reverse residue indexing of Section 4 remains in force:

$$e_t = u - 1 - r_t, \qquad q_t := [X^{e_t}]Q, \qquad c_t := [X^{e_t}]C \qquad (t \in \mathbb{Z}), \tag{212}$$

where a_t, e_t, q_t, and c_t are extended periodically with period u. Thus the coefficient sequences are pulled back from residue degrees to odd-phase order. A **boundary gap** means a cyclic gap between occupied sites in this pulled-back order; it is not a plateau length in residue order.

Since $t \mapsto e_t$ is a permutation, exactly K of the q_t are nonzero, while Theorem 4.5 gives

$$c_t \neq 0 \qquad \text{for every } t. \tag{213}$$

5.1 Boundary gaps force 3-adic growth

The factorization $Q = (2X^z - 3)C$ becomes a first-order recurrence when coefficients are indexed in odd-phase order.

Lemma 5.1 (Gap recurrence).

1. *For every $t \in \mathbb{Z}$,*

$$q_t = 2^{1+a_t} c_{t+1} - 3c_t. \tag{214}$$

2. *If* $q_{t+j} = 0$ *for* $0 \le j < \ell$, *where* $t \in \mathbb{Z}$ *and* $\ell \ge 0$, *then* $3^\ell \mid c_{t+\ell}$. *Since* $c_{t+\ell} \ne 0$, *this gives*

$$|c_{t+\ell}| \ge 3^\ell. \tag{215}$$

Proof. As in the proof of Theorem 4.4,

$$e_{t+1} + z = e_t + ua_t, \tag{216}$$

so multiplication by $2X^z$ carries the coefficient c_{t+1} to degree e_t with factor 2^{1+a_t}. Comparing the coefficient of X^{e_t} in $Q = (2X^z - 3)C$ gives (214).

For $\ell = 0$, both conclusions follow from $c_t \in \mathbb{Z} \setminus \{0\}$. For $\ell \ge 1$, iteration across the zero run yields

$$2^{\ell + \sum_{j=0}^{\ell-1} a_{t+j}} c_{t+\ell} = 3^\ell c_t. \tag{217}$$

Since powers of 2 and 3 are coprime, Euclid's lemma gives $3^\ell \mid c_{t+\ell}$. The final inequality follows because $c_{t+\ell}$ is a nonzero integer. □

Thus a zero run converts sparsity of Q into 3-adic divisibility in C; full support then converts that divisibility into an Archimedean lower bound.

Choose increasing integer lifts

$$s_1 < s_2 < \cdots < s_K < s_1 + u \tag{218}$$

of the odd-phase indices at which $q_t \ne 0$. Let

$$s_{K+1} := s_1 + u, \qquad m_i := s_{i+1} - s_i \quad (1 \le i \le K). \tag{219}$$

Thus the ith cyclic boundary gap has spacing m_i and contains the zero run $q_{s_i+1}, \dots, q_{s_{i+1}-1}$ of length $m_i - 1$. In particular,

$$m_i \ge 1, \qquad \sum_{i=1}^{K} m_i = u. \tag{220}$$

The longest gap immediately gives the estimate needed for the finite theorem.

Corollary 5.2 (Longest-gap growth).

One has

$$\|C\|_\infty \ge 3^{\lceil u/K \rceil - 1}. \tag{221}$$

Proof. Some spacing satisfies $m_i \ge \lceil u/K \rceil$. The intervening zero run has length $m_i - 1$, so Lemma 5.1 forces

$$|c_{s_{i+1}}| \ge 3^{m_i - 1}. \tag{222}$$

Taking the maximum over the coefficients of C proves the claim.

□

5.2 Bridge height and boundary mass

The relevant geometry is that of the residue-ordered height profile

$$H_0, H_1, \dots, H_{u-1}. \tag{223}$$

The bridge cannot reach a large height without creating many plateaux, and the resulting dyadic profile cannot produce boundary mass arbitrarily large relative to the plateau count.

Here boundary mass means the coefficient ℓ^1-norm $\|Q_L\|_1$. This is the useful measure because the reduced convolution in §5.3 bounds individual quotient coefficients by an ℓ^1-bound on the boundary and an ℓ^∞-bound on the other factor.

Theorem 5.3 (Height and boundary mass).

Every nonzero canonical bridge satisfies

$$M \le K - 2 \tag{224}$$

and

$$\|Q_L\|_1 \le (K+1)2^M - K \le (K+1)2^{K-2} - K. \tag{225}$$

Proof. Since

$$L_{t+1} - L_t \le a_t \le 1, \tag{226}$$

the bridge in odd-phase order must pass through every integer height $1, \dots, M$ on its way from 0 to its maximum. Residue reordering permutes the entries and therefore preserves the set of attained heights. Each of these M positive heights consequently occupies at least one plateau of the residue-ordered profile H.

The zero level contributes at least two further plateaux. To see this, let

$$\sigma(r) := r + z \pmod{u}. \tag{227}$$

If $r = r_t$, then $\sigma(r) = r_{t+1}$. A nonwrapping step $r + z < u$ has $a_t = 0$, so the bridge inequality implies

$$H_{\sigma(r)} \le H_r. \tag{228}$$

Suppose, contrary to the claim, that the zero entries of the linear profile form a single plateau. Since $H_0 = 0$ and H is not identically zero, there is an index $1 \le b < u$ such that

$$H_r = 0 \quad (0 \le r < b), \qquad H_r > 0 \quad (b \le r < u). \tag{229}$$

If $b \le z$, the nonwrapping step $0 \mapsto z$ moves from height 0 to a positive height. If $b > z$, the nonwrapping step $b - z \mapsto b$ does the same. Either case contradicts the bridge inequality. Thus the zero level has at least two plateaux, and

$$K \ge M + 2. \tag{230}$$

For the norm bound, recall that

$$Q_L(X) = (2d_0 - d_{u-1}) + \sum_{r=1}^{u-1} (d_r - d_{r-1}) X^{u-r}, \tag{231}$$

where

$$1 \le d_r \le 2^M, \qquad d_0 = 2^M. \tag{232}$$

There are exactly $K - 1$ nonzero interior differences. Each has absolute value at most $2^M - 1$, while the positive endpoint coefficient satisfies

$$2d_0 - d_{u-1} = 2^{M+1} - d_{u-1} \le 2^{M+1} - 1. \tag{233}$$

Consequently,

$$\begin{aligned} \|Q_L\|_1 &\le (2^{M+1} - 1) + (K - 1)(2^M - 1) \\ &= (K + 1)2^M - K. \end{aligned} \tag{234}$$

Substituting $M \le K - 2$ gives the final estimate. □

The height bound is geometric. The boundary-mass bound is dyadic. Together they convert the support count K into an explicit upper bound for the entire boundary mass.

5.3 Upper and lower quotient bounds

The explicit quotient formula of Theorem 4.1 bounds C from above in terms of Q. The following elementary operator estimate controls reduction modulo $X^u - 2$.

Lemma 5.4 (One-wrap convolution).

For $A, B \in \mathbb{Z}[X]_{<u}$,

$$\big\| [AB]_{X^u=2} \big\|_\infty \le 2 \|A\|_1 \|B\|_\infty. \tag{235}$$

Proof. Since $\deg(AB) < 2u$, each monomial wraps at most once. If

$$A = \sum_{i=0}^{u-1} a_i X^i, \qquad B = \sum_{j=0}^{u-1} b_j X^j, \tag{236}$$

then for $0 \le k < u$,

$$[X^k][AB]_{X^u=2} = \sum_{i+j=k} a_i b_j + 2 \sum_{i+j=k+u} a_i b_j. \tag{237}$$

For each fixed i, at most one of the two sums contains an admissible j. Taking absolute values gives the stated bound. □

Recall from §4.1 that $W_{n,u}$ is the geometric factor satisfying $\Psi_{n,u}W_{n,u} = D$. It reappears here because it converts $Q = \Psi_{n,u}C$ into an explicit formula for DC; reducing $W_{n,u}$ modulo $X^u - 2$ makes that formula amenable to coefficientwise bounds.

Let

$$\widetilde{W}_{n,u} := [W_{n,u}]_{X^u=2}. \tag{238}$$

Its coefficient at X^{r_t} is

$$\omega_t = 2^{t+A(t)}3^{u-1-t}. \tag{239}$$

Because $A(t) \le tz/u$,

$$\omega_t \le 2^{nt/u}3^{u-1-t}. \tag{240}$$

The ratio $2^{n/u}/3$ is greater than 1 because $D > 0$, so the right side increases with t. Hence, for $0 \le t < u$,

$$\omega_t \le 2^{n(u-1)/u} < \frac{2^n}{3}. \tag{241}$$

Therefore

$$\|\widetilde{W}_{n,u}\|_\infty < \frac{2^n}{3}. \tag{242}$$

The lower growth forced by a boundary gap can now be combined with the upper growth permitted by the dyadic boundary.

Theorem 5.5 (Finite plateau inequality).

Every integrally realizable canonical bridge satisfies

$$3^{\lceil u/K\rceil-1} < \frac{2}{3\delta}\,\|Q_L\|_1 \le \frac{2}{3\delta}\left((K+1)2^{K-2} - K\right). \tag{243}$$

Proof. The quotient formula gives

$$DC = [QW_{n,u}]_{X^u=2} = [Q\widetilde{W}_{n,u}]_{X^u=2}. \tag{244}$$

Lemma 5.4 and (242) imply

$$D\|C\|_\infty < \frac{2^{n+1}}{3}\|Q\|_1. \tag{245}$$

Since $D = 2^n\delta$,

$$\|C\|_\infty < \frac{2}{3\delta}\|Q\|_1. \tag{246}$$

Corollary 5.2 supplies the lower bound

$$\|C\|_\infty \geq 3^{\lceil u/K \rceil - 1}, \tag{247}$$

and Theorem 5.3 supplies the final upper bound for $\|Q\|_1$. □

5.4 Proof of Theorem 1.2

For $u \geq 1$, define

$$\mathscr{H}(u) := \max\left\{\log\left(\left(1 + \frac{2}{\log 3}\right) u\right) + 0.38,\ 10\right\}, \qquad \mathscr{E}(u) := 25.2\ \log 3\, \mathscr{H}(u)^2 + \log(u+1). \tag{248}$$

Proof. Theorem 2.3 gives the canonical bridge of the cycle's parity word, and Corollary 2.4 shows that it is integrally realizable. Proposition 2.2 gives

$$0 < z := n - u < u, \qquad D := 2^n - 3^u > 0. \tag{249}$$

Set

$$\delta := \frac{D}{2^n}, \qquad K := K(L). \tag{250}$$

Corollary A.2 gives

$$\log(1/\delta) \leq \log 2 + 25.2\ \log 3\, \mathscr{H}(u)^2, \tag{251}$$

while Theorem 5.5 gives

$$3^{\lceil u/K \rceil - 1} < \frac{2}{3\delta}(K+1)2^{K-2}. \tag{252}$$

Taking natural logarithms and using $\lceil u/K \rceil - 1 \geq u/K - 1$ gives

$$\left(\frac{u}{K} - 1\right) \log 3 < \log \frac{2}{3\delta} + \log(K+1) + (K-2)\log 2. \tag{253}$$

Adding $\log 3$ to both sides, and then applying the estimate from Corollary A.2, yields

$$\frac{u}{K} \log 3 < K \log 2 + 25.2\ \log 3\, \mathscr{H}(u)^2 + \log(K+1). \tag{254}$$

The plateau count satisfies $K \leq u$, so $\log(K+1) \leq \log(u+1)$. Multiplying by K therefore gives

$$u \log 3 < K\left(K \log 2 + \mathscr{E}(u)\right), \tag{255}$$

This is the explicit finite inequality underlying Theorem 1.2. Since $K > 0$, it is equivalent to the lower bound

$$K > \frac{\sqrt{\mathscr{E}(u)^2 + 4u \log 2 \log 3} - \mathscr{E}(u)}{2 \log 2}. \tag{256}$$

Moreover,

$$\sqrt{\mathscr{E}(u)^2 + 4u \log 2 \log 3} \ge 2\sqrt{u \log 2 \log 3}, \tag{257}$$

and hence

$$K > \sqrt{\log_2 3}\,\sqrt{u} - \frac{\mathscr{E}(u)}{2\log 2}. \tag{258}$$

Finally, there is an absolute constant $C > 0$ such that

$$\frac{\mathscr{E}(u)}{2\log 2} \le C(1 + \log u)^2 \qquad (u \ge 1), \tag{259}$$

because $\mathscr{H}(u) = O(1 + \log u)$ and $\mathscr{E}(u) = O((1 + \log u)^2)$ with absolute implied constants. Hence

$$K(L) \ge \sqrt{\log_2 3}\,\sqrt{u} - C(1 + \log u)^2, \tag{260}$$

uniformly over all hypothetical coprime positive cycles. This proves Theorem 1.2. □

Thus every hypothetical coprime cycle satisfies a finite plateau lower bound, with the leading constant $\sqrt{\log_2 3}$ and an explicit logarithmic loss.

6 Mechanical and one-swap exclusions

This section excludes the rooted mechanical word and every word obtained from it by one internal adjacent swap $01 \mapsto 10$ from encoding nontrivial positive integer Collatz cycles. In bridge coordinates, the mechanical word is $L = 0$ and the one-swap words are the singleton sequences $L^{(p)}$. These rooted objects are defined for every positive-denominator pair, but the exclusion argument then separates: the boundary-coordinate theory applies when $\gcd(n, u) = 1$, while a geometric cofactor replaces it when $\gcd(n, u) > 1$.

6.1 Rooted bridge candidates and the case split

Begin with a positive-denominator pair (n, u),

$$1 \le u < n, \qquad z := n - u, \qquad D := 2^n - 3^u > 0, \tag{261}$$

with no coprimality assumption. Set

$$A(t) := \left\lfloor \frac{tz}{u} \right\rfloor, \qquad a_t := A(t+1) - A(t). \tag{262}$$

Outside the coprime strict strip, the unique canonical-root correspondence of Theorem 2.3 is not available, and the increments a_t need not be binary. In this extended domain, a **rooted bridge candidate** is a sequence $L = (L_t)_{t=0}^{u}$ satisfying

$$L_0 = L_u = 0, \qquad L_t \ge 0 \quad (0 \le t \le u), \qquad L_{t+1} \le L_t + a_t \quad (0 \le t < u). \tag{263}$$

The reconstruction formulas still determine odd positions

$$j_t(L) = t + A(t) - L_t \tag{264}$$

and numerator

$$N(L) = \sum_{t=0}^{u-1} 2^{j_t(L)} 3^{u-1-t}. \tag{265}$$

The **mechanical bridge** is the zero sequence $L = 0$. For $1 \le p < u$, define the unit coordinate spike

$$L_t^{(p)} = \mathbf{1}_{\{t=p\}}. \tag{266}$$

The word *singleton* refers to the singleton support of L in odd-phase bridge coordinates. Call $L^{(p)}$ a **singleton bridge candidate** when it satisfies the reconstruction inequalities (263). When validity is understood, shorten the term to **singleton bridge**.

Let $\mathbf{a} = (a_t)_{t \bmod u}$, let $\boldsymbol{\gamma}$ be the reconstructed zero-gap sequence, and let $\mathbf{e}_q$ denote the qth standard basis vector in the cyclic zero-gap coordinate space. The zero-gap perturbation associated with $L^{(p)}$ is

$$\boldsymbol{\gamma} - \mathbf{a} = -\mathbf{e}_{p-1} + \mathbf{e}_p. \tag{267}$$

Thus one zero is moved from the gap following odd phase $p-1$ to the adjacent gap following odd phase p. This is narrower than an arbitrary single-zero relocation. For comparison, choose a root with $0 \le i < j < u$ and suppose the donor gap is nonempty. Then the perturbation

$$\boldsymbol{\gamma} - \mathbf{a} = -\mathbf{e}_i + \mathbf{e}_j \tag{268}$$

corresponds to the height-one interval bridge

$$L_t = \mathbf{1}_{\{i<t\le j\}}. \tag{269}$$

Only when $j = i + 1$ is this interval a singleton $L^{(p)}$. The exclusions proved below concern this adjacent-transfer subclass.

In the coprime strict strip, Theorem 2.3 identifies every singleton bridge candidate with the canonical bridge of its reconstructed parity word.

If either the mechanical bridge or a singleton bridge candidate encoded a nontrivial positive cycle, its displayed parity word would consist of repetitions of the least-period parity word. The displayed length and number of ones therefore scale by the same factor, so the inequalities of Proposition 2.2 place the displayed parameters in the strict strip

$$u < n < 2u, \qquad 0 < z < u. \tag{270}$$

The proof now divides by the value of $d := \gcd(n, u)$. In the coprime branch $d = 1$, residue order is a permutation and the boundary coordinates of Sections 3–5 apply directly. For displayed parameters with $d > 1$, residue order is not a permutation and those coordinates are unavailable. The mechanical bridge is handled by reduction to its primitive coprime block, whereas a noncoprime singleton is handled by the geometric cofactor of the repeated block. The two singleton branches are recombined in Theorem 6.5.

6.2 Boundary normal forms in the coprime strip

This subsection is confined to the coprime strict strip. Here residue order is a permutation, so the profile and boundary coordinates of Sections 3–5 are available and the zero and singleton bridges have completely explicit boundaries.

Lemma 6.1 (Minimal boundary normal forms).

Assume

$$\gcd(n,u) = 1, \qquad 0 < z < u, \qquad D > 0. \tag{271}$$

Then the following statements hold.

1. *For the mechanical bridge $L = 0$,*

$$M = 0, \qquad V_0(X) = 1 + X + \cdots + X^{u-1}, \qquad Q_0(X) = 1. \tag{272}$$

In particular,

$$K(0) = 1. \tag{273}$$

2. *The singleton sequence $L^{(p)}$ is a rooted bridge candidate if and only if*

$$a_{p-1} = 1. \tag{274}$$

Equivalently, let $r_p \in \{0, 1, \dots, u-1\}$ be the least residue of pz modulo u:

$$r_p \equiv pz \pmod{u}. \tag{275}$$

Then $L^{(p)}$ is a rooted bridge candidate if and only if

$$1 \le r_p < z. \tag{276}$$

For every such singleton,

$$H_r = \mathbf{1}_{\{r=r_p\}}, \tag{277}$$

and

$$Q_{L^{(p)}}(X) = 2 - X^{u-r_p} + X^{u-r_p-1}. \tag{278}$$

Consequently,

$$K(L^{(p)}) = 3, \qquad \|Q_{L^{(p)}}\|_1 = 4. \tag{279}$$

Proof. For $L = 0$, the residue-ordered height profile is identically zero, so $d_r = 1$ for every r. Hence

$$V_0(X) = \sum_{r=0}^{u-1} X^{u-1-r} \tag{280}$$

and

$$Q_0 = [(X-1)V_0]_{X^u=2} = [X^u - 1]_{X^u=2} = 1. \tag{281}$$

The height profile has one linear plateau.

For $L = L^{(p)}$, every bridge inequality is automatic except the one entering the exceptional height:

$$1 = L_p \le L_{p-1} + a_{p-1} = a_{p-1}. \tag{282}$$

Since $0 < z < u$, one has $a_{p-1} \in \{0, 1\}$, so the bridge condition is equivalent to $a_{p-1} = 1$. The residue recurrence

$$r_p = r_{p-1} + z - ua_{p-1} \tag{283}$$

shows that a wrap at this step is equivalent to $0 \le r_p < z$. Coprimality and $1 \le p < u$ exclude $r_p = 0$, proving the residue criterion.

Whenever this criterion holds, $L^{(p)}$ is a valid rooted bridge candidate and hence, by Theorem 2.3, the canonical bridge of its reconstructed parity word. The residue permutation therefore places the unique height 1 at r_p. Thus $M = 1$ and

$$d_r = 2 - \mathbf{1}_{\{r=r_p\}}. \tag{284}$$

Because $1 \le r_p < z < u$, the exceptional residue is neither endpoint of the linear profile. Applying the boundary operator gives

$$Q_{L^{(p)}}(X) = 2 - X^{u-r_p} + X^{u-r_p-1}. \tag{285}$$

The height profile consists of an initial zero plateau, the singleton height-1 plateau, and a final zero plateau. The support and norm assertions follow. □

6.3 Mechanical-bridge exclusion

The mechanical bridge can now be treated over the full positive-denominator domain of §6.1. Its unit boundary excludes the coprime branch directly, while the noncoprime branch reduces to the primitive mechanical block.

For the zero bridge,

$$j_t(0) = t + \left\lfloor \frac{t(n-u)}{u} \right\rfloor = \left\lfloor \frac{tn}{u} \right\rfloor. \tag{286}$$

Hence its one-based odd positions are

$$1 + \left\lfloor \frac{tn}{u} \right\rfloor \qquad (0 \le t < u), \tag{287}$$

which is the upper-Christoffel position formula for Knight's high-cycle parity class, up to the choice of cyclic root [5]. The same Christoffel class also appears as the extremizer of the rotation-invariant minimum cycle numerator studied by Fernández and Ibáñez [6, Theorem 7.3].

Proposition 6.2 (Mechanical-bridge exclusion).

For every pair

$$1 \le u < n, \qquad 2^n - 3^u > 0, \tag{288}$$

the mechanical bridge encodes no nontrivial positive integer cycle. In the coprime strip this follows immediately from $Q_0 = 1$; for noncoprime parameters, the mechanical word reduces to a repetition of its primitive Christoffel block.

Proof. The following boundary-theoretic argument is self-contained and includes the reduction of the noncoprime case.

Suppose that the mechanical bridge encodes a nontrivial positive cycle. The least-period reduction of §6.1 and Proposition 2.2 place its displayed parameters in the strict strip

$$u < n < 2u. \tag{289}$$

Write

$$d := \gcd(n, u), \qquad (n, u, z) = d(n_0, u_0, z_0). \tag{290}$$

If $d = 1$, then $D > 1$ in the strict strip. Lemma 6.1 gives $Q_0 = 1$, whereas Theorem 3.4 would require

$$Q_0(\xi) = 0 \pmod{D} \tag{291}$$

for integral realization. This is impossible.

Suppose $d > 1$. The reduced parameters satisfy

$$\gcd(n_0, u_0) = 1, \qquad 0 < z_0 < u_0, \qquad 2^{n_0} - 3^{u_0} > 1. \tag{292}$$

Indeed, the strict-strip inequalities descend after division by d, while $D > 0$ implies $2^{n_0} > 3^{u_0}$; the final strict inequality is the coprime strict-strip observation from §2.1. Writing $t = ju_0 + s$ with $0 \leq j < d$ and $0 \leq s < u_0$ now gives

$$j_{ju_0+s}(0) = jn_0 + \left\lfloor \frac{sn_0}{u_0} \right\rfloor. \tag{293}$$

Thus the mechanical parity word is a d-fold repetition

$$w_0 = v^d \tag{294}$$

of the reduced mechanical word v with parameters (n_0, u_0). Let

$$f := T_v. \tag{295}$$

$$f(x) = \alpha x + \beta, \qquad \alpha = \frac{3^{u_0}}{2^{n_0}} \in (0, 1). \tag{296}$$

If the repeated word fixed a positive integer x, then $f^d(x) = x$. The affine iteration identity

$$f^d(x) - x = (1 + \alpha + \cdots + \alpha^{d-1})(f(x) - x) \tag{297}$$

would imply $f(x) = x$. The same integer would therefore realize the reduced coprime mechanical word, whose boundary is $Q_0 = 1$ modulo $2^{n_0} - 3^{u_0} > 1$. The coprime argument gives a contradiction. □

6.4 A geometric cofactor excludes noncoprime singletons

Now assume $\gcd(n, u) > 1$. The coprime residue-order boundary is not used in this subsection. Instead, the mechanical numerator repeats with a geometric cofactor; a singleton perturbation changes only one summand, and that local change cannot absorb the cofactor.

Proposition 6.3 (Noncoprime singleton exclusion).

Suppose

$$d := \gcd(n, u) > 1, \qquad D = 2^n - 3^u > 0. \tag{298}$$

Then no singleton bridge candidate $L^{(p)}$ is integrally realizable.

Proof. Write

$$(n, u, z) = d(n_0, u_0, z_0), \qquad \mathsf{U} := 2^{n_0}, \qquad \mathsf{V} := 3^{u_0}. \tag{299}$$

Then

$$D = (\mathsf{U} - \mathsf{V})S_d, \tag{300}$$

where

$$S_d := \sum_{j=0}^{d-1} \mathsf{U}^j \mathsf{V}^{d-1-j}. \tag{301}$$

Because $D > 0$, one has $\mathsf{U} > \mathsf{V}$; together with $d > 1$, this gives $S_d > 1$.

For the mechanical bridge define

$$\omega_t := 2^{t+A(t)}3^{u-1-t}, \tag{302}$$

so that

$$N(0) = \sum_{t=0}^{u-1} \omega_t. \tag{303}$$

If $t = ju_0 + s$ with $0 \le s < u_0$, then

$$A(ju_0 + s) = jz_0 + \left\lfloor \frac{sz_0}{u_0} \right\rfloor. \tag{304}$$

Consequently,

$$\omega_{ju_0+s} = \mathsf{U}^j \mathsf{V}^{d-1-j} \omega_s^{(0)}, \tag{305}$$

where

$$\omega_s^{(0)} := 2^{s+\lfloor sz_0/u_0 \rfloor}3^{u_0-1-s}. \tag{306}$$

Summing first over j gives

$$N(0) = S_d \sum_{s=0}^{u_0-1} \omega_s^{(0)}. \tag{307}$$

A singleton bridge lowers only the pth odd position by one. Therefore

$$N(L^{(p)}) = N(0) - \frac{\omega_p}{2}. \tag{308}$$

Since $p \geq 1$ and $A(p) \geq 0$, the quantity $\omega_p/2$ is an integer. If $D \mid N(L^{(p)})$, then $S_d \mid N(L^{(p)})$ because $S_d \mid D$. Since $S_d \mid N(0)$, it follows that

$$S_d \mid \frac{\omega_p}{2}. \tag{309}$$

This is impossible. Indeed,

$$S_d \equiv 1 \pmod 2 \tag{310}$$

and

$$S_d \equiv \mathsf{U}^{d-1} \not\equiv 0 \pmod 3. \tag{311}$$

Thus S_d is coprime to 6. On the other hand,

$$\frac{\omega_p}{2} = 2^{p+A(p)-1} 3^{u-1-p} \tag{312}$$

is a positive monomial in 2 and 3. Its only divisor coprime to 6 is 1, contradicting $S_d > 1$. □

6.5 Coprime singleton exclusion

With the noncoprime singleton branch excluded, it remains to treat the coprime strict strip. By Lemma 6.1, the singleton boundary has three occupied coefficients and ℓ^1 norm 4. The finite plateau inequality therefore forces the normalized denominator to be exponentially small in u. Laurent's lower bound for $n \log 2 - u \log 3$ prevents this above the explicit threshold $u \geq 10^6$.

Proposition 6.4 (Coprime singleton exclusion above the explicit threshold).

Assume

$$\gcd(n, u) = 1, \qquad u < n < 2u, \qquad D = 2^n - 3^u > 0, \qquad u \geq 10^6. \tag{313}$$

Then no singleton bridge candidate is integrally realizable.

Proof. Suppose that $L^{(p)}$ is integrally realizable. By Lemma 6.1,

$$K(L^{(p)}) = 3, \qquad \|Q_{L^{(p)}}\|_1 = 4. \tag{314}$$

The finite plateau inequality, Theorem 5.5, gives

$$3^{\lceil u/3\rceil-1} < \frac{8}{3\delta}, \qquad \delta := \frac{D}{2^n}. \tag{315}$$

Here $D > 0$ gives

$$\Lambda = n\log 2 - u\log 3 > 0, \qquad \delta = \frac{D}{2^n} = 1 - e^{-\Lambda}. \tag{316}$$

Proposition A.3 applies to positive integers n, u satisfying $u < n < 2u$, $\Lambda > 0$, and $u \geq 10^6$, and gives

$$3^{\lceil u/3\rceil-1} \geq \frac{8}{3\delta}. \tag{317}$$

This contradicts (315). Hence $L^{(p)}$ is not integrally realizable. □

The explicit range follows from one published cycle-length estimate. In the shortcut-map convention used here, with one iterate and one parity symbol counted per step, Eliahou proved [15] that every nontrivial positive cycle has total least period

$$n \geq 17{,}087{,}915. \tag{318}$$

The stronger Hercher–Bařina odd-member bound quoted in the introduction is unnecessary here; Eliahou's estimate is sufficient because the present argument needs only $u \geq 10^6$.

Combined with $n < 2u$, this gives

$$u \geq 8{,}543{,}958. \tag{319}$$

6.6 Mechanical and one-swap exclusion

The final theorem returns once more to arbitrary positive-denominator pairs and combines the two mutually exclusive gcd branches.

Theorem 6.5 (Singleton-bridge exclusion).

For every positive-denominator pair (n, u),

$$1 \leq u < n, \qquad 2^n - 3^u > 0, \tag{320}$$

no singleton bridge candidate encodes a nontrivial positive integer Collatz cycle.

Proof. Suppose that a singleton bridge candidate encodes a nontrivial positive cycle, and let $d := \gcd(n, u)$. If $d = 1$, Proposition 2.5 makes the displayed period n the least period, and Proposition 2.2 gives $u < n < 2u$. Eliahou's bound then implies

$$u \geq 8{,}543{,}958 > 10^6. \tag{321}$$

Proposition 6.4 gives a contradiction. If $d > 1$, Proposition 6.3 gives a contradiction directly. These two cases are exhaustive. □

Corollary 6.6 (Mechanical and one-swap exclusion).

For every pair

$$1 \le u < n, \qquad 2^n - 3^u > 0, \tag{322}$$

let w_{mech} be the rooted mechanical parity word of length n and weight u. Neither w_{mech} nor any rooted word obtained from it by one internal adjacent swap

$$01 \mapsto 10 \tag{323}$$

encodes a nontrivial positive integer Collatz cycle.

Proof. Proposition 6.2 excludes w_{mech}. An indicated swap moves one noninitial odd symbol from its mechanical position $j_p(0)$ to $j_p(0)-1$ and leaves all other odd positions fixed. Its rooted bridge sequence is therefore $L_t = \mathbf{1}_{\{t=p\}}$, and the occurrence of 01 ensures that the reconstructed zero gaps are nonnegative. Theorem 6.5 excludes every such singleton bridge. □

A singleton bridge has one occupied bridge-height coordinate and therefore transfers one zero between adjacent mechanical gaps. In the noncoprime case, the resulting monomial perturbation cannot absorb the geometric cofactor of $2^n - 3^u$; in the coprime case, its three-term boundary creates a gap incompatible with the available lower bound for $n \log 2 - u \log 3$. Arbitrary single-zero relocations correspond instead to height-one interval bridges and remain outside these two exclusions.

7 Discussion and outlook

The preceding sections establish an exact boundary representation for the coprime prescribed-parity problem and derive from it a quantitative lower bound on plateau complexity. The mechanical and singleton bridge families are also excluded throughout the positive-denominator domain, using separate arguments in the coprime and noncoprime branches. This final section situates the plateau invariant among existing complexity measures, identifies the principal losses in the present estimates, and describes the obstacles to extending the boundary theory beyond coprimality.

7.1 Plateau complexity among cycle-complexity invariants

The representation theorem of §1.2 summarizes the structural passage developed in Sections 2–4: a canonically rooted parity word determines a bridge, residue reordering turns its heights into a positive dyadic profile, and the boundary operator compresses that profile to $K(L)$ occupied coefficients. Integral realizability is then equivalent to divisibility of this sparse boundary by the distinguished factor $\Psi_{n,u}$, while the quotient square retains a positive polynomial that recovers the odd orbit. For a least-period realization, the associated boundary quotient has full support. The resulting tension is therefore not merely between a small polynomial and a large modulus; it is between a sparse geometric boundary and a dense arithmetic quotient.

These are genuine constraints on parity-pattern structure, not merely on an auxiliary polynomial: Theorem 1.1 shows that Q_L determines the canonical bridge and hence the original cyclic parity class. The constraints are nevertheless expressed after residue reordering, rather than directly in the visible 0-1 word. Translating $K(L)$ into native-order features such as runs, blocks, or local transitions remains open.

The finite plateau inequality of Theorem 5.5 quantifies that tension. With

$$K := K(L), \qquad \delta := \frac{2^n - 3^u}{2^n}, \tag{324}$$

its logarithmic consequence can be written

$$u \log 3 \le K^2 \log 2 + K \log(1/\delta) + O(K \log(K+1)). \tag{325}$$

The implied constant is absolute.

Theorem 1.2 controls the denominator term through Laurent's estimate and gives every hypothetical coprime cycle the pointwise lower bound $K(L) \ge \sqrt{\log_2 3}\,\sqrt{u} - C(1+\log u)^2$ for an absolute constant $C > 0$. This is a deterministic forbidden-region result, not a distributional statement: the present work neither determines the typical size of $K(L)$ among legal canonical bridges nor asserts that the square-root scale is sharp. The argument leaves open the entire range between this critical scale and linear plateau complexity.

Plateau count belongs to a broader family of cycle-complexity measures, but it records different information. Brox's descent number counts odd-to-odd steps requiring at least two divisions by 2 [11]. The m-cycle count used by Simons and collaborators records alternating monotone blocks, equivalently local odd minima [12–14]. These closely related measures are defined in the native order of the parity word or orbit. By contrast, $K(L)$ counts constant intervals only after canonical rooting and residue reordering. No identity or comparison inequality between these native-order measures and $K(L)$ is asserted. Nevertheless, the common principle is visible:

Low complexity concentrates the combinatorial change at relatively few sites, leaving long structured intervals between them. In the boundary coordinate, the occupied boundary sites correspond to $K(L) - 1$ interior profile transitions and one endpoint coefficient. Pulling these sites back to odd-phase order produces the complementary cyclic gaps that force the 3-adic growth used in Section 5. A comparison theorem relating native-order block complexity to residue-order plateau complexity would make this arithmetic obstruction visible directly in ordinary parity-word geometry.

7.2 Where the present size squeeze loses information

Both sides of the finite plateau inequality are uniform estimates, and both discard structure that an extremal argument could retain.

The boundary-mass estimate

$$\|Q_L\|_1 \le (K+1)2^{K-2} - K \tag{326}$$

uses the height bound $M \le K - 2$, the number of nonzero profile changes, and the largest possible dyadic jump. It does not use the full bridge inequalities after residue reordering or the nesting of the height superlevel sets

$$\{r : H_r \ge h\}. \tag{327}$$

A boundary near this envelope would require nearly maximal height and nearly maximal dyadic change at many interior profile transitions simultaneously. It is not known whether a canonical bridge can realize such a configuration.

The quotient estimate loses different information. Within an unoccupied boundary gap, the recurrence from §5.1 forces a 3-adic cascade. At an occupied site, however, the full relation

$$q_t = 2^{1+a_t} c_{t+1} - 3c_t \tag{328}$$

couples the cascades on its two sides. The present lower bound treats the gaps independently. It also uses none of the additional facts that $C_L = [(X-1)Y_L]_{X^u=2}$, that Y_L is coefficientwise positive, that its exact 2-adic valuations recover the bridge heights, and that its normalized coefficients are distinct odd orbit entries. Thus the upper and lower estimates may have incompatible near-extremizers even when each estimate is individually sharp over a larger class of vectors.

The losses isolate three open problems.

1. **Boundary envelope.** Determine the sharp maximum of $\|Q_L\|_1$ among legal canonical bridges with prescribed $K(L)$, or prove a uniform improvement over the exponent of 2^K in the present envelope. Such a result must use compatibility among the reordered bridge inequalities and the nested height superlevel sets, rather than optimizing arbitrary profile changes independently.
2. **Coupled boundary gaps.** Derive a lower quotient bound that retains the relations across occupied boundary sites instead of treating the complementary gaps independently. The central issue is whether the positivity, exact 2-adic valuations, and orbit interpretation of the quotient prevent the independent gap estimates from being simultaneously sharp.
3. **Critical plateau scale.** Decide whether integrally realizable canonical bridges can satisfy

$$K(L) = \left(\sqrt{\log_2 3} + o(1)\right) \sqrt{u}, \tag{329}$$

or whether their normalized plateau count must remain uniformly above the constant supplied by Theorem 1.2.

Near equality in the final bound would require simultaneous near-equality in the gap distribution, the height-to-plateau estimate, the boundary-mass estimate, and the convolution bound. Classifying their joint approximate extremizers may therefore be more informative than sharpening any one inequality in isolation.

The scope distinction made in Section 6 supplies a smaller intermediate test family. A singleton bridge candidate is defined by $L_t = \mathbf{1}_{\{t=p\}}$ for some $1 \le p < u$ and moves one zero only between adjacent gaps. A nonwrapping relocation to a nonadjacent later gap gives a height-one interval bridge candidate

$$L_t = \mathbf{1}_{\{i<t\le j\}}, \qquad 0 \le i < j < u, \qquad j \ge i+2. \tag{330}$$

Only candidates satisfying the bridge legality conditions of Theorem 2.3 belong to the canonical family; relocations crossing the canonical cut require a corresponding rerooting and are not represented by this nonwrapping display. For a legal height-one interval bridge, the numerator perturbation contains several monomials, while residue reordering can create more than three occupied boundary coefficients. The two singleton proofs therefore do not extend formally. A uniform exclusion of height-one interval bridges is a concrete intermediate problem between the singleton theorem and the general boundary-envelope problem.

7.3 Beyond coprimality

The coprime boundary theory depends essentially on the permutation

$$t \mapsto tz \pmod{u}. \tag{331}$$

If

$$d := \gcd(u,z) > 1, \qquad u = du_0, \qquad z = dz_0, \tag{332}$$

then the map visits only u_0 residues. Writing $t = ju_0 + s$ gives

$$tz \bmod u = d(sz_0 \bmod u_0), \tag{333}$$

independently of the sheet index j. A one-component residue profile therefore collapses the d bridge heights

$$L_{ju_0+s} \qquad (0 \le j < d) \tag{334}$$

lying above the same reduced residue. The injective boundary reconstruction of Theorem 3.3 cannot survive that collapse unchanged.

The natural raw replacement is the d-sheeted array

$$H_{j,s} := L_{ju_0+s}, \qquad 0 \le j < d, \quad 0 \le s < u_0. \tag{335}$$

Any multicomponent replacement for the coprime boundary theory would need:

1. injective reconstruction of the full bridge together with its root data;
2. an exact integral-factorization criterion that retains both the reduced factor $2^{n_0} - 3^{u_0}$ and the geometric cofactor;
3. an orbit-recovery quotient with a support or noncancellation theorem strong enough to sustain gap estimates; and
4. a complexity invariant stable under interactions among the sheets.

Section 6 shows that the geometric cofactor alone excludes a singleton bridge in the noncoprime branch. For a general bridge—even for a height-one interval bridge—several numerator terms may interact across different sheets, and their sum may acquire factors absent from every individual term. Retaining the missing coordinates is therefore only the first step. A multicomponent plateau argument would additionally require a quantitative noncancellation theorem across sheets.

The main contribution of the present work is therefore an exact coordinate system in which geometric simplicity and arithmetic realizability meet. The canonical boundary records deviation from the mechanical word. Its principal-ideal factorization records integrality. The associated positive profile quotient Y_L retains the orbit itself. The plateau theorem shows that these roles cannot be separated: a boundary with too few occupied coefficients cannot support the arithmetic density required by a full-period integral orbit.

In the coprime boundary coordinate, the identity $Q_0 = 1$ excludes the mechanical bridge; primitive-block reduction extends that exclusion to the noncoprime branch. The singleton theorem then excludes every valid singleton bridge candidate on the full positive-denominator domain. Height-one interval bridges are not uniformly excluded. The sharp general plateau scale is unknown. Further progress may therefore come from a finer classification of the boundary factorizations actually produced by legal bridges—their joint extremal geometry, quotient signs and valuations, and sheet interactions—rather than from the undifferentiated cycle numerator alone.

A Explicit logarithmic estimates

This appendix records the two-logarithm specialization used in Sections 5 and 6 and verifies the explicit threshold in the coprime singleton argument. All logarithms in this appendix are natural.

A.1 Laurent's two-logarithm estimate in the cycle strip

Let

$$\Lambda := n \log 2 - u \log 3. \tag{336}$$

For positive-denominator parameters, $\Lambda > 0$.

Theorem A.1 (Specialization of Laurent).

Let n, u be positive integers satisfying

$$u < n < 2u, \qquad \Lambda > 0. \tag{337}$$

Set

$$c := 1 + \frac{2}{\log 3} \tag{338}$$

and

$$\mathcal{H}(u) := \max\left\{\log(cu) + 0.38,\ 10\right\}. \tag{339}$$

Then

$$\log \Lambda \geq -25.2\ \log 3\ \mathcal{H}(u)^2. \tag{340}$$

Proof. Apply the explicit estimate for a linear form in two logarithms [7, Corollary 2 and Table 1] with

$$\alpha_1 = 3, \qquad \alpha_2 = 2, \qquad b_1 = u, \qquad b_2 = n, \tag{341}$$

field-degree parameter $D_K = 1$, auxiliary parameter $m = 10$, and

$$\log A_1 = \log 3, \qquad \log A_2 = 1. \tag{342}$$

The positive rationals 2 and 3 are multiplicatively independent. Their absolute logarithmic heights are $\log 2$ and $\log 3$, so the choices of A_1, A_2 in (342) satisfy Laurent's height conditions. His coefficient parameter obeys

$$\begin{aligned} b' &= \frac{u}{\log A_2} + \frac{n}{\log A_1} \\ &= u + \frac{n}{\log 3} \\ &\leq \left(1 + \frac{2}{\log 3}\right) u = cu. \end{aligned} \tag{343}$$

The constant in Laurent's table for $m = 10$ is $C_2(10) = 25.2$, yielding (340). □

A.2 Consequence for the normalized denominator

Let

$$\delta := \frac{2^n - 3^u}{2^n} = 1 - e^{-\Lambda}. \tag{344}$$

Corollary A.2.

Under the hypotheses of Theorem A.1,

$$\log(1/\delta) \le \log 2 + 25.2\ \log 3\, \mathscr{H}(u)^2. \tag{345}$$

In particular,

$$\log(1/\delta) = O((\log u)^2) \tag{346}$$

uniformly in the cycle-relevant strip.

Proof. If $0 < \Lambda \le 1$, then

$$1 - e^{-\Lambda} \ge \frac{\Lambda}{2}, \tag{347}$$

so

$$\log(1/\delta) \le \log 2 - \log \Lambda. \tag{348}$$

Theorem A.1 gives the result. If $\Lambda > 1$, then

$$\delta > 1 - e^{-1} > \frac{1}{2}, \tag{349}$$

and (345) is immediate. □

A.3 Explicit incompatibility at the singleton scale

Proposition A.3 (Singleton scale versus Laurent).

Let n, u be positive integers satisfying

$$u < n < 2u, \qquad \Lambda = n \log 2 - u \log 3 > 0, \qquad u \ge 10^6, \tag{350}$$

and

$$\delta = 1 - e^{-\Lambda}. \tag{351}$$

Then

$$3^{\lceil u/3 \rceil - 1} \ge \frac{8}{3\delta}. \tag{352}$$

Proof. Let

$$\ell := \left\lceil \frac{u}{3} \right\rceil - 1. \tag{353}$$

Then

$$\ell \geq \frac{u-3}{3}. \tag{354}$$

Suppose, to the contrary, that

$$3^{\ell} < \frac{8}{3\delta}. \tag{355}$$

Then

$$\delta < \frac{8}{3}\, 3^{-\ell} < 4 \cdot 3^{-\ell}. \tag{356}$$

Since $u \geq 10^6$, one has $\ell \geq 3$, so $\delta < 1/2$. Therefore

$$\Lambda = -\log(1-\delta) < 2\delta < 8 \cdot 3^{-\ell}, \tag{357}$$

and hence

$$\log \Lambda < \log 8 - \ell \log 3 \leq \log 8 - \frac{u-3}{3} \log 3. \tag{358}$$

Let

$$G(u) := \frac{u-3}{3} \log 3 - 25.2\, \log 3\, \mathscr{H}(u)^2 - \log 8. \tag{359}$$

The upper bound (358) is incompatible with Theorem A.1 whenever $G(u) > 0$. We verify this for every $u \geq 10^6$. Set $U := 10^6$. The elementary estimates

$$\frac{2}{3} < \log 2 < \frac{7}{10}, \qquad \log 3 > 1, \qquad c < 3, \tag{360}$$

along with

$$2^{19} < U, \qquad 3U < 2^{22}, \tag{361}$$

give

$$\log(cU) + \frac{19}{50} > 19 \cdot \frac{2}{3} + \frac{19}{50} > 10 \tag{362}$$

and

$$\log(cU) + \frac{19}{50} < 22 \cdot \frac{7}{10} + \frac{19}{50} < 16. \tag{363}$$

Thus the logarithmic branch of $\mathscr{H}$ is active at U and

$$10 < \mathscr{H}(U) < 16. \tag{364}$$

Since $8 < 9$,

$$\frac{\log 8}{\log 3} < 2. \tag{365}$$

It follows that

$$\begin{aligned} \frac{G(U)}{\log 3} &> \frac{U-3}{3} - \frac{126}{5}16^2 - 2 \\ &= \frac{4{,}903{,}187}{15} > 0. \end{aligned} \tag{366}$$

For $u \geq U$, the logarithmic branch remains active, so

$$\mathcal{H}'(u) = \frac{1}{u} \tag{367}$$

and

$$G'(u) = \log 3 \left(\frac{1}{3} - \frac{252}{5}\frac{\mathcal{H}(u)}{u}\right). \tag{368}$$

Moreover,

$$\left(\frac{\mathcal{H}(u)}{u}\right)' = \frac{1-\mathcal{H}(u)}{u^2} < 0. \tag{369}$$

Thus

$$\frac{\mathcal{H}(u)}{u} \leq \frac{\mathcal{H}(U)}{U} < \frac{16}{10^6}, \tag{370}$$

and

$$\frac{252}{5}\frac{\mathcal{H}(u)}{u} < \frac{4032}{5{,}000{,}000} < \frac{1}{3}. \tag{371}$$

Hence $G'(u) > 0$ for all $u \geq U$. Since $G(U) > 0$, one has $G(u) > 0$ throughout the stated range, contradicting the simultaneous Laurent lower bound and the assumed singleton-scale upper bound. □

B Research provenance and AI disclosure

Development was conducted primarily through OpenAI Codex integrated with Visual Studio Code, with more limited use of Anthropic Claude Code and other language-model tools.

Work often proceeded iteratively across systems. Claude Code commonly supported textual explanations and section drafts; Codex commonly generated and reviewed mathematical arguments, developed code, ran computational experiments, and helped prepare requests for theoretical advances; and OpenAI GPT models generated further theoretical proposals. Some prompts were drafted by Codex and revised by the author, while others were drafted directly by the author. The author set and redirected the research program, reviewed, selected, corrected, and integrated the resulting work, and has recently focused especially on making the results intelligible, coherent, and human-readable in a presentation consistent with his own explanatory voice.

The author accepts sole responsibility for the presentation, final claims, and citations. No AI system is an author. At present, the author has reviewed and checked most results through Section 3 with access to the manuscript and supporting materials. He can explain the principal structures, motivations, and argument flow but does not claim complete unaided proof reconstruction. Results after Section 3 have not yet received the same level of author review. AI-assisted criticism and agreement among models are not independent human verification. This manuscript has not been peer reviewed.

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